

Differential Equations and Vector Calculus

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April 24, 2025

1 Vector Calculus

1.1 Double Integration:

1.

$$\text{Volume} = \lim_{n \rightarrow \infty} \left[\sum_{j=1}^n \sum_{i=1}^n f(x_i, y_j) dA_{ij} \right] = \iint_R f(x, y) dA$$

2. When $f(x, y) = 1$, $\text{Area} = \iint_R dA$

1.2 Fubini's Theorem:

1. If $f(x, y)$ is continuous on a rectangle region R : $a \leq x \leq b$, $c \leq y \leq d$, then

$$\iint_R f(x, y) dA = \int_a^b \int_c^d f(x, y) dy dx = \int_c^d \int_a^b f(x, y) dx dy$$

2. **Question:** If $f(x, y) = 100 - 6x^2y$, then find $\iint_R f(x, y) dA$ for R : $0 \leq x \leq 2$, $-1 \leq y \leq 1$. **Solution:**

$$\begin{aligned} \iint_R (100 - 6x^2y) dA &= \int_{y=-1}^1 \left[\int_{x=0}^2 (100 - 6x^2y) dx \right] dy \\ &= \int_{-1}^1 (100 - 6x^2y) \Big|_0^2 dy = \int_{-1}^1 (200 - 16y) dy = 400 \end{aligned}$$

1.2.1 Fubini's Theorem (Stronger Form):

Let $f(x, y)$ be continuous on a region R .

1. If R is defined by $a \leq x \leq b$, $g_1(x) \leq y \leq g_2(x)$, then

$$\iint_R f(x, y) dA = \int_a^b \left[\int_{g_1(x)}^{g_2(x)} f(x, y) dy \right] dx$$

2. If R is defined by $h_1(y) \leq x \leq h_2(y)$, $c \leq y \leq d$, then

$$\iint_R f(x, y) dA = \int_c^d \left[\int_{h_1(y)}^{h_2(y)} f(x, y) dx \right] dy$$

1.3 Polar Curves:

$$\iint_R f(x, y) dA = \int_{\theta=\alpha}^{\theta=\beta} \left[\int_{r=f_2(\theta)}^{r=f_1(\theta)} f(r, \theta) dr \right] d\theta$$

We have $x^2 + y^2 = r^2 \implies |r| = \sqrt{x^2 + y^2}$ and $\theta = \tan^{-1} \left(\frac{y}{x} \right)$

1.3.1 Jacobian to Convert Systems:

- 1.

$$\iint_R F(x, y) dx dy = \iint_{R^*} F(u, v) \left| J \left(\frac{x, y}{u, v} \right) \right| du dv$$

where

$$J \left(\frac{x, y}{u, v} \right) = \begin{vmatrix} \partial x / \partial u & \partial x / \partial v \\ \partial y / \partial u & \partial y / \partial v \end{vmatrix}$$

2. For converting $(x, y) \rightarrow (r, \theta)$
we know, $x = r \cos \theta$, $y = r \sin \theta$

$$x_r = \cos \theta, x_\theta = -r \sin \theta, y_r = \sin \theta, y_\theta = r \cos \theta$$

$$J \left(\frac{x, y}{r, \theta} \right) = \begin{vmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{vmatrix} = r$$

Hence,

$$\boxed{\iint_R F(x, y) dx dy = \iint_{R^*} F(r, \theta) r dr d\theta}$$

1. **Question:** Evaluate: $\int_0^\infty \int_0^\infty e^{-(x^2+y^2)} dx dy$

Solution: Let $x^2 + y^2 = r^2 \implies dx dy = r dr d\theta$

$$\begin{aligned} \int_0^\infty \int_0^\infty e^{-(x^2+y^2)} dx dy &= \int_0^{\pi/2} \int_0^\infty e^{-r^2} r dr d\theta \\ &= \int_0^{\pi/2} \left[-\frac{1}{2} e^{-r^2} \right]_0^\infty d\theta = \int_0^{\pi/2} \frac{1}{2} d\theta = \frac{\pi}{4} \end{aligned}$$

2. **Question:** Evaluate: $I = \int_0^\infty e^{-x^2} dx$

Solution: We know,

$$I^2 = \int_0^\infty \int_0^\infty e^{-(x^2+y^2)} dx dy = \frac{\pi}{4} \implies I = \frac{\sqrt{\pi}}{2}$$

3. **Question:** Evaluate: $\iint_R \cos(x+2y)\sin(x-y)dxdy$ bounded by region R: $y = 0$, $y = x$, and $x + 2y = 8$.

Solution:

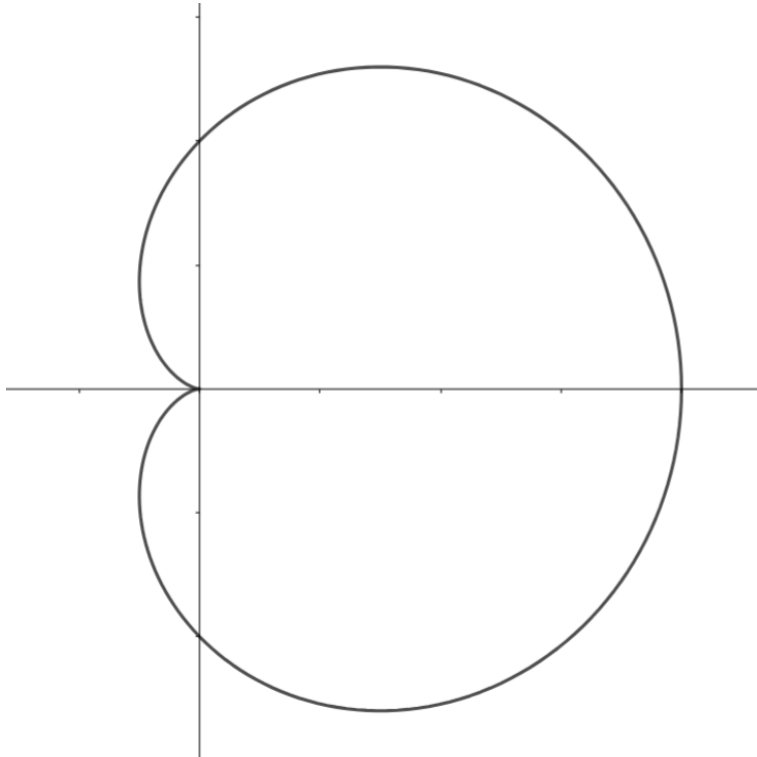
$$\text{Let } x + 2y = u, x - y = v \implies 3x = u + 2v, 3y = u - v$$

$$J\left(\frac{x,y}{u,v}\right) = \begin{vmatrix} 1/3 & 2/3 \\ 1/3 & -1/3 \end{vmatrix} = -1/3$$

$$\begin{aligned} \therefore \iint_R \cos(x+2y)\sin(x-y)dxdy &= -1/3 \iint_{R^*} \cos(u)\sin(v)dvdu \\ &= \int_0^8 \left[\int_0^u \frac{-1}{3} \cos u \cdot \sin v dv \right] du = 1/3 \cdot \int_0^8 \cos u \cdot \sin u du \\ &= \int_0^8 \frac{1}{6} \sin(2u) du = \frac{1}{12} [\cos(16) - 1] \end{aligned}$$

1.3.2 Cardioids:

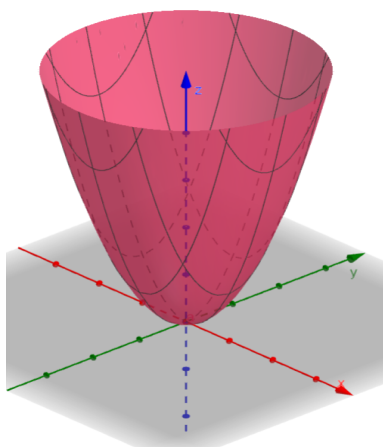
$$r = a(1 + \cos\theta)$$



1.4 3D Shapes:

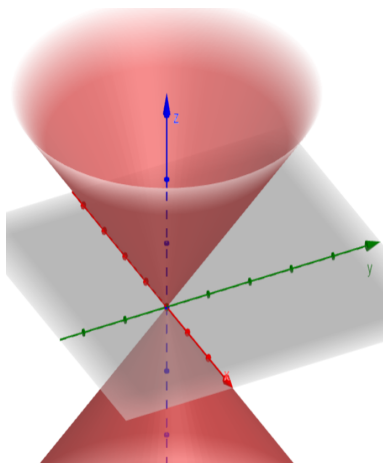
1.4.1 Paraboloid:

$$z = x^2 + y^2$$



1.4.2 Cone:

$$z^2 = x^2 + y^2$$



1.5 Triple Integration:

$$\iiint_V f(x, y, z) dV = \lim_{n \rightarrow \infty} \left[\sum_{i=1}^n f(x_i, y_i, z_i) dV_i \right]$$

If $f(x, y, z) = 1$, then $\text{Volume} = \iiint_V dV$ where $dV = dx dy dz$

Question: Find the volume of the region enclosed between the surfaces $z = x^2 + 3y^2$ and $z = 8 - x^2 - y^2$.

Solution: Intersection,

$$x^2 + 3y^2 = 8 - x^2 - y^2 \implies 2x^2 + 4y^2 = 8 \implies y = \sqrt{\frac{4-x^2}{2}}$$

$$\begin{aligned} V &= 4 \int_0^2 \int_0^{\sqrt{\frac{4-x^2}{2}}} \int_{x^2+3y^2}^{8-x^2-y^2} dz dy dx = 4 \int_0^2 \int_0^{\sqrt{\frac{4-x^2}{2}}} (8-2x^2-4y^2) dy dx \\ &= 4 \int_0^2 \left[8y - 2x^2y - \frac{4}{3}y^3 \right] \Big|_0^{\sqrt{\frac{4-x^2}{2}}} dx = 4 \int_0^2 \left[(8-2x^2)\sqrt{\frac{4-x^2}{2}} - \frac{4}{3} \left(\frac{4-x^2}{2} \right)^{3/2} \right] dx \\ &= 2\sqrt{2} \int_0^2 \left[2(4-x^2)^{3/2} - \frac{4}{3}(4-x^2)^{3/2} \right] dx = \frac{8\sqrt{2}}{3} \int_0^2 (4-x^2)^{3/2} dx \end{aligned}$$

Now, consider $I = \int_0^2 (4-x^2)^{3/2} dx$

Let $x = 2\sin\theta \implies dx = 2\cos\theta d\theta$

$$\begin{aligned} I &= \int_0^{\pi/2} (4-4\sin^2\theta)^{3/2} (2\cos\theta d\theta) = 16 \int_0^{\pi/2} \cos^4\theta d\theta \\ &= 16 \cdot \frac{\Gamma(\frac{1}{2}) \Gamma(\frac{5}{2})}{2\Gamma(3)} = \frac{16}{2 \cdot 2} \cdot \sqrt{\pi} \left(\frac{3}{2} \frac{1}{2} \sqrt{\pi} \right) = 3\pi \end{aligned}$$

Hence,

$$V = \frac{8\sqrt{2}}{3} \cdot 3\pi = 8\sqrt{2}\pi$$

1.6 Moments and Center of Mass:

1.6.1 First Moments:

1. For a 3-D Solid:

Mass (M) = $\iiint \rho dV$ where ρ is the density.

$$M_{yz} = \iiint x \cdot \rho dV, M_{xz} = \iiint y \cdot \rho dV, M_{xy} = \iiint z \cdot \rho dV$$

$$\bar{x} = \frac{M_{yz}}{M}, \bar{y} = \frac{M_{xz}}{M}, \bar{z} = \frac{M_{xy}}{M}$$

2. For a 2-D Plate:

Mass (M) = $\iint \sigma dA$ where σ is the density.

$$M_y = \iint x \cdot \sigma dA, M_x = \iint y \cdot \sigma dA$$

$$\bar{x} = \frac{M_y}{M}, \bar{y} = \frac{M_x}{M}$$

1.6.2 Second Moments:

1. For a 3-D Solid:

- (a) About x-axis, $I_x = \iiint (y^2 + z^2) \cdot \rho dV$
- (b) About y-axis, $I_y = \iiint (x^2 + z^2) \cdot \rho dV$
- (c) About z-axis, $I_z = \iiint (x^2 + y^2) \cdot \rho dV$
- (d) About a line L, $I_L = \iiint r^2(x, y, z) \cdot \rho dV$

2. For a 2-D Plate:

- (a) About x-axis, $I_x = \iint y^2 \cdot \sigma dA$
- (b) About y-axis, $I_y = \iint x^2 \cdot \sigma dA$
- (c) About a line L, $I_L = \iint r^2(x, y) \cdot \sigma dA$
- (d) About origin, $I_o = \iint (x^2 + y^2) \cdot \sigma dA$

1.7 Cylindrical Coordinates:

- 1. Convert $(x, y, z) \leftrightarrow (r, \theta, z)$
- 2. Let $x = r\cos\theta$, $y = r\sin\theta$, $z = z$
- 3. As z is not affected, it is same as polar conversion $\implies \text{---J---} = r$.
- 4. Hence,

$$\iiint_V f(x, y, z) dV = \iiint_{V^*} F(r, \theta, z) r dr d\theta dz$$

- 5. **Question:** Find the volume of the paraboloid $x^2 + y^2 = 4z$ and $z = 4$.
Solution:

$$z = \frac{1}{4}(x^2 + y^2) = 4 \implies x^2 + y^2 = 16$$

$$\therefore V = \int_0^4 \int_0^{\sqrt{16-x^2}} \int_{\frac{1}{4}(x^2+y^2)}^4 dz dy dx$$

Converting into cylindrical coordinates, let $x = r\cos\theta$, $y = r\sin\theta$, $z = z$

$$V = 4 \int_0^{\pi/2} \int_0^4 \int_{r^2/4}^4 r \cdot dz dr d\theta = 4 \int_0^{\pi/2} \int_0^4 \left[4r - \frac{r^3}{4} \right] dr d\theta$$

$$\therefore V = 2\pi \left[2r^2 - \frac{r^4}{16} \right] \Big|_0^4 = 32\pi$$

1.8 Spherical Coordinates:

1. Convert $(x, y, z) \leftrightarrow (\rho, \theta, \phi)$
2. where ρ is the distance from origin, θ is the angle from x-axis, and ϕ is the angle from z-axis.
3. Now, $\sin\phi = r/\rho$ and $\cos\phi = z/\rho$
 $\implies r = \rho\sin\phi$ and $z = \rho\cos\phi$

4.

$$\therefore x = \rho \sin\phi \cos\theta, y = \rho \sin\phi \sin\theta, z = \rho \cos\phi$$

5. Now,

$$J \left(\frac{x, y, z}{\rho, \theta, \phi} \right) = \begin{vmatrix} \sin\phi \cos\theta & \rho \cos\phi \cos\theta & \rho \sin\phi \cos\theta \\ \sin\phi \sin\theta & \rho \cos\phi \sin\theta & \rho \sin\phi \sin\theta \\ \cos\phi & -\rho \sin\phi & 0 \end{vmatrix} = \rho^2 \sin\phi$$

6. Hence,

$$\iiint_V f(x, y, z) dV = \iiint_{V^*} F(\rho, \theta, \phi) \rho^2 \sin\phi d\rho d\theta d\phi$$

7. **Notes:**

- (a) $\rho > 0, 0 \leq \phi \leq \pi, 0 \leq \theta \leq 2\pi$
- (b) $\rho = a$ is a sphere of radius a.
- (c) $\phi = \alpha$ is a cone with vertex at origin and angle α .
- (d) $\theta = \beta$ is a plane passing through z-axis and making angle β with x-axis.
- (e) $\rho = z$ is a cylinder along z-axis.

8. **Question:**

$$\text{Evaluate: } \int_0^1 \int_0^{\sqrt{1-x^2}} \int_0^{\sqrt{1-x^2-y^2}} \frac{dx dy dz}{\sqrt{1-x^2-y^2-z^2}}$$

Solution: Clearly, it is a sphere of radius 1 in the first octant. Hence, converting to spherical coordinates, $0 \leq \rho \leq 1, 0 \leq \theta \leq \pi/2, 0 \leq \phi \leq \pi/2$.

$$\begin{aligned} V &= \int_0^{\pi/2} \int_0^{\pi/2} \int_0^1 \frac{\rho^2 \sin\phi}{\sqrt{1-\rho^2}} d\rho d\phi d\theta = \frac{\pi}{2} \int_0^{\pi/2} \int_0^1 \frac{\rho^2 \sin\phi}{\sqrt{1-\rho^2}} d\rho d\phi \\ &= \frac{\pi}{2} \int_0^1 \left[\frac{\rho^2 \cos\phi}{\sqrt{1-\rho^2}} \right]_{\pi/2}^0 d\rho = \frac{\pi}{2} \int_0^1 \frac{\rho^2}{\sqrt{1-\rho^2}} d\rho \end{aligned}$$

$$\text{Let } u^2 = 1 - \rho^2 \implies u du = -\rho d\rho \text{ and } \rho = \sqrt{1-u^2}$$

$$\therefore V = \frac{\pi}{2} \int_0^1 \frac{\sqrt{1-u^2}}{u} u du = \frac{\pi}{2} \int_0^1 \sqrt{1-u^2} du = \frac{\pi}{8}$$

9. **Question:** Find the volume of the icecream cone cut from the solid sphere $\rho \leq 1$ by the cone $\phi = \pi/3$.

Solution:

$$V = \int_{\theta=0}^{2\pi} \int_{\phi=0}^{\pi/3} \int_{\rho=0}^1 \rho^2 \sin\phi d\rho d\phi d\theta = 2\pi \int_0^{\pi/3} \left. \frac{\rho^3}{3} \right|_0^1 \sin\phi d\phi$$

$$\therefore V = \frac{2\pi}{3} [-\cos\phi]_0^{\pi/3} = -\frac{2\pi}{3} \left(\frac{1}{2} - 1 \right) = \frac{\pi}{3}$$

1.8.1 Jacobian in 3 Variables:

1.

$$J \left(\frac{x, y, z}{u, v, w} \right) = \begin{vmatrix} \partial x / \partial u & \partial x / \partial v & \partial x / \partial w \\ \partial y / \partial u & \partial y / \partial v & \partial y / \partial w \\ \partial z / \partial u & \partial z / \partial v & \partial z / \partial w \end{vmatrix}$$

2.

$$J \left(\frac{x, y, z}{u, v, w} \right) \times J \left(\frac{u, v, w}{x, y, z} \right) = 1$$

3. **Question:**

$$\text{Evaluate: } \int_0^3 \int_0^4 \int_{y/2}^{y/2+1} \left(\frac{2x-y}{2} - \frac{z}{3} \right) dx dy dz$$

Solution: Let $u = 2x - y, v = y/2, w = z/3$

$$J \left(\frac{u, v, w}{x, y, z} \right) = \begin{vmatrix} 1/2 & -1/2 & 0 \\ 0 & 1/2 & 0 \\ 0 & 0 & 1/3 \end{vmatrix} = 1/6 \implies J = 6$$

$$\therefore V = \int_{u=0}^1 \int_{v=0}^2 \int_{w=0}^1 6(u+w) dw dv du = 12$$

1.9 Vector Valued Functions:

- Let $r(t) = f(t)\hat{i} + g(t)\hat{j} + h(t)\hat{k}$ be a vector function with domain D and let L be a vector. We say that r has a limit L as t approaches t_0 if:

$$\lim_{t \rightarrow t_0} r(t) = L \iff \lim_{t \rightarrow t_0} f(t) = L_x, \lim_{t \rightarrow t_0} g(t) = L_y, \lim_{t \rightarrow t_0} h(t) = L_z$$

- A vector function is continuous at a point $t = t_0$ if $\lim_{t \rightarrow t_0} r(t) = r(t_0)$

- Velocity: $v = \frac{dr}{dt}$, Acceleration: $a = \frac{dv}{dt} = \frac{d^2r}{dt^2}$

- The indefinite integral of r with respect to t is the set of all anti-derivatives of r: $\int r(t)dt = R(t)dt + c$

5. **Arc Length:** The length of the smooth curve $r(t) = x(t)\hat{i} + y(t)\hat{j} + z(t)\hat{k}$ that is traced entirely as t increases from $t = a$ to $t = b$ is:

$$L = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} dt$$

Hence, $L = \int_a^b v(t) dt$

6. **Unit Tangent Vector:**

$$T = \frac{v}{|v|} \text{ where } v = \frac{dr}{dt}$$

7. **Curvature:**

$$\kappa = \left| \frac{dT}{ds} \right| = \left| \frac{dT}{dt} \right| \left| \frac{ds}{dt} \right|^{-1} \implies \kappa = \frac{1}{|v|} \left| \frac{dT}{dt} \right|$$

8. At a point where $\kappa \neq 0$, the principal unit normal for a smooth curve is

$$N = \frac{1}{\kappa} \frac{dT}{ds} \frac{dT/dt}{|dT/dt|}$$

9. **Binormal Vector:**

$$B = T \times N$$

10. **Acceleration:**

$$a = \frac{d^2s}{dt^2} T + \kappa \left(\frac{ds}{dt} \right)^2 N = a_T T + a_N N$$

11. **Curvature in terms of Acceleration:**

$$\kappa = \frac{|v \times a|}{|v|^3}$$

12. **Torsion:**

$$\frac{1}{|v \times a|^2} \begin{vmatrix} x' & y' & z' \\ x'' & y'' & z'' \\ x''' & y''' & z''' \end{vmatrix}$$

1.9.1 Properties of Vectors:

1. $\vec{u} + \vec{v} = \vec{v} + \vec{u}$
2. $\vec{u} + (\vec{v} + \vec{w}) = (\vec{u} + \vec{v}) + \vec{w}$
3. $\vec{u} + \vec{0} = \vec{u}$

4. $\vec{u} + (-\vec{u}) = \vec{0}$
5. $\vec{0} \cdot \vec{u} = \vec{u}$
6. $1\vec{u} = \vec{u}$
7. $c(d\vec{u}) = (cd)\vec{u}$
8. $(c + d)\vec{u} = c\vec{u} + d\vec{u}$

1.9.2 Unit Vector:

A unit vector in the direction of vector $\vec{u}$ is denoted by $\hat{u}$ and is given by:

$$\hat{u} = \frac{\vec{u}}{|\vec{u}|}, |\hat{u}| = 1$$

Some Special Unit Vectors:

$$\hat{i} = \langle 1, 0, 0 \rangle, \hat{j} = \langle 0, 1, 0 \rangle, \hat{k} = \langle 0, 0, 1 \rangle$$

For a vector $\vec{v} = \langle v_1, v_2, v_3 \rangle$, it can be represented as:

$$\vec{v} = v_1\hat{i} + v_2\hat{j} + v_3\hat{k}$$

1.9.3 Dot Product:

The dot product of two vectors $\vec{a}(\langle a_1, a_2, a_3 \rangle)$ and $\vec{b}(\langle b_1, b_2, b_3 \rangle)$ is:

$$\vec{a} \cdot \vec{b} = a_1b_1 + a_2b_2 + a_3b_3$$

$$\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}|\cos(\theta)$$

where θ is the angle between the two vectors.

1.9.4 Cross Product:

The cross product of two vectors $\vec{a}(\langle a_1, a_2, a_3 \rangle)$ and $\vec{b}(\langle b_1, b_2, b_3 \rangle)$ is:

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

$$\vec{a} \times \vec{b} = (a_2b_3 - a_3b_2)\hat{i} + (a_3b_1 - a_1b_3)\hat{j} + (a_1b_2 - a_2b_1)\hat{k}$$

$$\vec{a} \times \vec{b} = |\vec{a}||\vec{b}|\sin(\theta)\hat{n}$$

where θ is the angle between the two vectors and $\hat{n}$ is the unit vector perpendicular to the plane containing $\vec{a}$ and $\vec{b}$.

1.10 The Nabla Operator:

$$\nabla = \hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z}$$

1.10.1 Gradient (∇):

The gradient of a scalar function $f(x, y, z)$ is a vector function ∇f defined as:

$$\nabla f = \hat{i} \frac{\partial f}{\partial x} + \hat{j} \frac{\partial f}{\partial y} + \hat{k} \frac{\partial f}{\partial z}$$

1.10.2 Divergence ($\nabla \cdot$):

Divergence is a scalar function defined as:

$$\nabla \cdot \vec{F} = \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z}$$

1.10.3 Curl ($\nabla \times$):

The curl is a vector function defined as:

$$\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_x & F_y & F_z \end{vmatrix}$$
$$\nabla \times F = \hat{i} \left(\frac{\partial F_z}{\partial y} - \frac{\partial F_y}{\partial z} \right) - \hat{j} \left(\frac{\partial F_z}{\partial x} - \frac{\partial F_x}{\partial z} \right) + \hat{k} \left(\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y} \right)$$

1.10.4 Properties of Divergence and Curl:

1. If $\nabla \cdot \vec{F} = 0$, then $\vec{F}$ is said to be "solenoidal vector".
2. If $\nabla \times \vec{F} = 0$, then $\vec{F}$ is said to be "irrotational vector".
3. $\nabla \times \nabla f = 0 \implies \text{grad}(f)$ is always irrotational.

1.11 Line Integrals:

$$\text{If } I = \int_C F(x, y, z) ds = \lim_{n \rightarrow \infty} \sum_{i=1}^n F(x_i, y_i, z_i) \Delta s_i$$

Parametrizing, let $x = x(t)$, $y = y(t)$, $z = z(t)$ for $a \leq t \leq b$.

$$\int_C F(x, y, z) ds = \int_a^b F(x(t), y(t), z(t)) \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} dt$$
$$\therefore I = \int_a^b F(x(t), y(t), z(t)) |v(t)| dt$$

1. **Question:** Integrate $f(x) = x - 3y^2 + z$ over the line segment C joining the origin to the point (1, 1, 1).

Solution: Let $x = t, y = t, z = t. \implies |v| = \sqrt{3}$

$$\begin{aligned}\int_0^1 f(t, t, t)|v(t)| dt &= \int_0^1 (t - 3t^2 + t)(\sqrt{3})dt = \int_0^1 (2t - 3t^2)(\sqrt{3})dt \\ &= (\sqrt{3}) [t^2 - t^3]_0^1 = (\sqrt{3})(1 - 1) = 0.\end{aligned}$$

2. **Question:** Integrate $f(x) = x - 3y^2 + z$ over the union of line segment C_1 (joining the origin to the point (1, 1, 0)) and C_2 (further joining to (1, 1, 1)).

Solution: For $C_1, x = t, y = t, z = 0 \implies |v_1| = \sqrt{2}$

$$\begin{aligned}\int_0^1 f(t, t, 0)(\sqrt{2})dt &= (\sqrt{2}) \int_0^1 (t - 3t^2)dt \\ &= (\sqrt{2}) \left[\frac{t^2}{2} - t^3 \right]_0^1 = (\sqrt{2}) \frac{1}{2} - 1 = -\frac{1}{\sqrt{2}}.\end{aligned}$$

For $C_2, x = 1, y = 1, z = t \implies |v_2| = 1$

$$\begin{aligned}\int_0^1 f(1, 1, t)dt &= \int_0^1 (1 - 3 + t)dt = \int_0^1 (-2 + t)dt \\ &= -2t + \frac{t^2}{2} \Big|_0^1 = -2 + \frac{1}{2} = -\frac{3}{2}.\end{aligned}$$

Hence, the total integral is $-\frac{1}{\sqrt{2}} - \frac{3}{2}$.

1.11.1 Mass and Moment for Springs, Wires, Thin Rods lying on a smooth curve C in space:

1. **Mass:** $M = \int_C \rho ds$ where $\rho = \rho(x, y, z)$ is the density.

2. **First Moments:**

$$M_{yz} = \int_C x \rho ds, \quad M_{xz} = \int_C y \rho ds, \quad M_{xy} = \int_C z \rho ds$$

3. **Moment of Inertias:**

$$I_x = \int_C (y^2 + z^2) \rho ds, \quad I_y = \int_C (x^2 + z^2) \rho ds, \quad I_z = \int_C (x^2 + y^2) \rho ds$$

4. **Question:** A slender metal arch, denser at the bottom than the top lies along the semicircle $y^2 + z^2 = 1, z \geq 0$ in the y-z plane. Find the center of the arch's mass if the density at point (x, y, z) is $\rho(x, y, z) = 2 - z$.

Solution: As it lies in y-z plane, $\bar{x} = 0$.

Similarly, due to symmetry, $\bar{y} = 0$.

Now, parametrizing, let $x = 0$, $y = \sin(t)$, $z = \cos(t) \implies |v| = 1$.

$$\begin{aligned} M &= \int_{-\pi/2}^{\pi/2} (2 - z(t))(1)dt = \int_{-\pi/2}^{\pi/2} (2 - \cos(t))dt \\ &= [2t - \sin(t)]_{-\pi/2}^{\pi/2} = 2\pi - 2 = 2(\pi - 1). \end{aligned}$$

Now,

$$\begin{aligned} M_{xy} &= \int_C z \rho ds = \int_{-\pi/2}^{\pi/2} \cos(t)(2 - \cos(t))(1)dt \\ &= \int_{-\pi/2}^{\pi/2} (2\cos(t) - \cos^2(t))dt = 4 - \frac{\pi}{2} \end{aligned}$$

Hence,

$$\bar{z} = \frac{8 - \pi}{4(\pi - 1)}, \quad \bar{x} = 0, \quad \bar{y} = 0$$

1.11.2 Work Done:

Let C be a smooth curve parameterized by $\vec{r}(t)$ ($a \leq t \leq b$) and $\vec{F}$ be a continuous force field over a region containing C , then the work done in moving an object from point A ($\vec{r}(a)$) to B ($\vec{r}(b)$) along C is:

$$W = \int_C \vec{F} \cdot \vec{T} ds = \int_a^b \vec{F}(\vec{r}(t)) \frac{d\vec{r}}{dt} dt$$

Let $\vec{F} = M\hat{i} + N\hat{j} + P\hat{k}$, and $\vec{r}(t) = x(t)\hat{i} + y(t)\hat{j} + z(t)\hat{k}$, then:

$$W = \int_C (Mdx + Ndy + Pdz)$$

1.11.3 Flow Integral:

If $\vec{r}(t)$ parameterizes a smooth curve C in the domain of a continuous velocity field $\vec{F}$, then the flow along the curve from A to B is:

$$\text{Flow} = \int_C \vec{F} \cdot \vec{T} ds$$

The integral is called as a "flow integral". If the curve starts and ends at the same point such that $A = B$, then flow is called "circulation around the curve".

1.11.4 Flux:

If C is a smooth, simple, closed curve in the domain of a continuous vector field $\vec{F} = M(x, y)\hat{i} + N(x, y)\hat{j}$ in the plane and if $\hat{n}$ is the outward pointing unit normal vector on C, the flux of $\vec{F}$ across C is:

$$\phi = \int_C \vec{F} \cdot \hat{n} ds$$

$$\text{Now, } \hat{n} = \vec{T} \times \hat{k} = \frac{dy}{ds}\hat{i} - \frac{dx}{ds}\hat{j}$$

$$\text{Hence, } \phi = \int_C (Mdy - Ndx)$$

1.11.5 Questions:

1. **Question:** Find work done by a force field $\vec{F} = (y - x^2)\hat{i} + (z - y^2)\hat{j} + (x - z^2)\hat{k}$ along $r(t) = t\hat{i} + t^2\hat{j} + t^3\hat{k}$ from (0,0,0) to (1,1,1)

Solution:

$$\begin{aligned} \vec{r} &= \langle t, t^2, t^3 \rangle \implies \vec{v} = \langle 1, 2t, 3t^2 \rangle \\ \implies \vec{F} &= (t^2 - t^2)\hat{i} + (t^3 - t^4)\hat{j} + (t - t^6)\hat{k} \\ \vec{F} \cdot \vec{v} &= 2t(t^3 - t^4) + 3t^2(t - t^6) = 3t^3 + 2t^4 - 2t^5 - 3t^7 \\ \therefore W &= \int_0^1 (\vec{F} \cdot \vec{v}) dt = \frac{3}{4} + \frac{2}{5} - \frac{1}{3} - \frac{3}{8} = \frac{29}{60} \end{aligned}$$

2. **Question:** Find the circulation of force field $\vec{F} = (x - y)\hat{i} + x\hat{j}$ along $r(t) = \cos(t)\hat{i} - \sin(t)\hat{j}$ ($0 \leq t \leq 2\pi$)

Solution:

$$\begin{aligned} \vec{r} &= \langle \cos(t), \sin(t) \rangle \implies \vec{v} = \langle -\sin(t), \cos(t) \rangle \\ \implies \vec{F} &= (\cos(t) - \sin(t))\hat{i} + \cos(t)\hat{j} \\ \vec{F} \cdot \vec{v} &= -\sin t \cos t + \sin^2 t + \cos^2 t = 1 - \sin t \cos t \\ \therefore C &= \int_0^{2\pi} \vec{F} \cdot \vec{v} dt = \int_0^{2\pi} (1 - \sin t \cos t) dt = 2\pi \end{aligned}$$

3. **Question:** Find flux for the vector field $\vec{F} = (x - y)\hat{i} + x\hat{j}$ along $x^2 + y^2 = 1$ in the x-y plane.

Solution: Let $x = \cos(t)$, $y = \sin(t) \implies \vec{F} = (\cos(t) - \sin(t))\hat{i} + \cos(t)\hat{j}$

$$\begin{aligned} \therefore \phi &= \int (Mdy - Ndx) = \int [\cos(t)(\cos(t) - \sin(t)) - \sin(t)\cos(t)] \\ &= \int_0^{2\pi} [\cos^2 t - \sin(2t)] dt = \pi \end{aligned}$$

1.11.6 Path Independence:

1. Let $\vec{F}$ be a vector field defined on an open region D in space and suppose that for any two points A and B in D, the line integral $\int_C \vec{F} \cdot d\vec{r}$ along a path C from A to B in D is the same over any path from A to B, then the integral is "path independent" in D and the field $\vec{F}$ is conservative on D.
2. If $\vec{F}$ is a vector field defined in D and $\vec{F} = \nabla f$ for some scalar function f, then f is called a "potential function" of $\vec{F}$.

$$\int_C \vec{F} \cdot d\vec{r} = \int_A^B \nabla f \cdot d\vec{r} = f(B) - f(A)$$

1.11.7 Fundamental Theorem of Line Integrals:

Let C be a smooth curve joining the points A and B in the plane or in a space, parameterized by $\vec{r}(t)$. Let f be a differentiable function with a continuous gradient vector ($\vec{F} = \nabla f$) on the domain D containing C, then:

$$\int_C \vec{F} \cdot d\vec{r} = f(B) - f(A)$$

1.11.8 Conservative Fields as Gradient Fields:

Let $\vec{F} = M\hat{i} + N\hat{j} + P\hat{k}$ be a vector field whose components are continuous throughout an open connected region D in the space. Then, $\vec{F}$ is conservative iff $\vec{F}$ is a gradient field ∇f for a differential function f.

Hence, $\vec{F}$ is conservative iff: $\vec{F} = \nabla f$

1.11.9 Component Test for Conservative Fields:

Let $\vec{F} = M(x, y, z)\hat{i} + N(x, y, z)\hat{j} + P(x, y, z)\hat{k}$ be a field on an open, simply connected domain whose component functions have continuous first partial derivatives. $\vec{F}$ is conservative iff:

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}, \quad \frac{\partial N}{\partial z} = \frac{\partial P}{\partial x}, \quad \frac{\partial P}{\partial y} = \frac{\partial M}{\partial z}$$

Question: Show that $\vec{F} = (e^x \cos y + yz)\hat{i} + (xz - e^x \sin y)\hat{j} + (xy + z)\hat{k}$ is conservative and find its potential function.

Solution: Let $M = e^x \cos y + yz, N = xz - e^x \sin y, P = xy + z$.

$$\frac{\partial M}{\partial y} = -e^x \sin y + z = \frac{\partial N}{\partial x}, \quad \frac{\partial N}{\partial z} = x = \frac{\partial P}{\partial y}, \quad \frac{\partial P}{\partial x} = y = \frac{\partial M}{\partial z}$$

Hence, $\vec{F}$ is conservative.

$$\text{Now, let } f(x, y, z) = \int M dx = \int (e^x \cos y + yz) dx = e^x \cos y + xyz + g(y, z)$$

$$\text{Now, } \frac{\partial f}{\partial y} = -e^x \sin y + xz + \frac{\partial g}{\partial y} = -e^x \sin y + xz \implies \frac{\partial g}{\partial y} = 0$$

Hence, $f(x, y, z) = e^x \cos y + xyz + g(z)$

$$\text{Now, } \frac{\partial f}{\partial z} = xy + \frac{\partial g}{\partial z} = xy + z \implies \frac{\partial g}{\partial z} = z$$

$$\therefore f(x, y, z) = e^x \cos y + xyz + \frac{z^2}{2} + c$$

1.11.10 Component Test for Exactness of $Mdx + Ndy + Pdz$:

1. Any expression $M(x, y, z)dx + N(x, y, z)dy + P(x, y, z)dz$ in a differential form is called "exact" on a domain D in space if:

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}, \quad \frac{\partial N}{\partial z} = \frac{\partial P}{\partial x}, \quad \frac{\partial P}{\partial y} = \frac{\partial M}{\partial z}$$

for some scalar function f throughout D .

2. This is equivalent to saying that the field $\vec{F}$ is conservative.

1.12 Green's Theorem:

1.12.1 Circulation - Curl / Tangential Form:

Let C be a piecewise, smooth, simple closed curve enclosing a region R in a plane. Let $\vec{F} = M\hat{i} + N\hat{j}$ be a vector field with M and N having continuous first partial derivatives in an open region containing R . Then the counterclockwise circulation of $\vec{F}$ around C equals to the double integral of $(\vec{\nabla} \times \vec{F}) \cdot \hat{k}$ over R .

$$\therefore \oint_C \vec{F} \cdot \vec{T} ds = \oint_C (Mdx + Ndy) = \iint_R (\vec{\nabla} \times \vec{F}) \cdot \hat{k} dA$$

Now,

$$\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ M & N & 0 \end{vmatrix} \implies (\vec{\nabla} \times \vec{F}) \cdot \hat{k} = \frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}$$

Hence,

$$\oint_C (Mdx + Ndy) = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dA$$

1.12.2 Flux Form:

$$\oint_C \vec{F} \cdot \hat{n} ds = \oint_C (Mdy - Ndx) = \iint_R \left(\frac{\partial M}{\partial x} + \frac{\partial N}{\partial y} \right) dA$$

1. **Question:** Evaluate the line integral $\oint_C (xydy - y^2dx)$ where C is a square cut from the first quadrant by the line $x = 1$ and $y = 1$.

Solution: $M = xy, N = y^2 \implies M_x = y, N_y = 2y \implies M_x + N_y = 3y$

$$\oint_C (xydy - y^2dx) = \iint_R 3y dxdy = \int_0^1 \int_0^1 3y dxdy = \frac{3}{2}$$

2. Calculate outward flux of $\vec{F} = 2e^{xy}\hat{i} + y^3\hat{j}$ across the square bounded by the lines $x = \pm 1$ and $y = \pm 1$.

Solution: $M = 2e^{xy}, N = y^3 \implies M_x = 2ye^{xy}, N_y = 3y^2 \implies M_x + N_y = 2ye^{xy} + 3y^2$

$$\begin{aligned} \oint_C (2e^{xy}dy - y^3dx) &= \iint_R (2ye^{xy} + 3y^2) dxdy = \int_{-1}^1 \int_{-1}^1 (2ye^{xy} + 3y^2) dxdy \\ &= \int_{-1}^1 [2e^{xy} + y^3] \Big|_{-1}^1 dy = \int_{-1}^1 (2e^y + y^3 - 2e^{-y} - y^3) dy = 4 \end{aligned}$$

1.13 Surface Integrals:

1.13.1 Surface Area:

1. **Parametric Form:** The area of the smooth surface

$$\vec{r}(u, v) = f(u, v)\hat{i} + g(u, v)\hat{j} + h(u, v)\hat{k}$$

is:

$$A = \iint_R |\vec{r}_u \times \vec{r}_v| dA = \iint_R |\vec{r}_u \times \vec{r}_v| dudv = \iint_R d\sigma$$

2. **Implicit Form:** The area of the surface $F(x, y, z) = c$ over a closed and bounded region R is:

$$A = \iint_R \frac{|\vec{\nabla} f|}{|\vec{\nabla} f \cdot \hat{p}|} dA$$

where $\hat{p}$ is the outward normal vector to the surface.

3. **Relation between parametric and implicit forms:**

For implicit form, $F(x, y, z) = c$.

Let $x = u, y = v, z = h(u, v) \implies \vec{r}(u, v) = u\hat{i} + v\hat{j} + h(u, v)\hat{k}$

$$\vec{r}_u = \hat{i} + \frac{\partial h}{\partial u} \hat{k}, \quad \vec{r}_v = \hat{j} + \frac{\partial h}{\partial v} \hat{k}$$

$$\implies \vec{r}_u = \hat{i} - \frac{F_x}{F_z} \hat{k}, \quad \vec{r}_v = \hat{j} - \frac{F_y}{F_z} \hat{k}$$

$$\therefore |\vec{r}_u \times \vec{r}_v| = \left| \frac{F_x}{F_z} \hat{i} + \frac{F_y}{F_z} \hat{j} + \hat{k} \right| = \left| \frac{F_x + F_y + F_z}{F_z} \right|$$

$$\therefore |\vec{r}_u \times \vec{r}_v| = \frac{|\vec{\nabla} f|}{|\vec{\nabla} f \cdot \hat{p}|}$$

4. **Explicit Form:** For a function $z = f(x, y)$ over the region R in xy plane, the surface area is:

$$\iint_{R(x,y)} \sqrt{f_x^2 + f_y^2 + 1} \, dxdy$$

Also,

$$\iint_{R(y,z)} \sqrt{1 + f_y^2 + f_z^2} \, dydz, \quad \iint_{R(x,z)} \sqrt{f_x^2 + 1 + f_z^2} \, dxdz$$

1.13.2 Surface Integrals of Scalar Functions:

1. For a scalar function $G(x, y, z)$ the surface integral is:

$$SI = \iint_S G(x, y, z) d\sigma$$

2. **Parametric Form:** $\vec{r}(u, v) = f(u, v)\hat{i} + g(u, v)\hat{j} + h(u, v)\hat{k}$

$$\iint_S G(x, y, z) d\sigma = \iint_R G(f(u, v), g(u, v), h(u, v)) |\vec{r}_u \times \vec{r}_v| \, dudv$$

3. **Implicit Form:** $F(x, y, z) = C$

$$\iint_S G(x, y, z) d\sigma = \iint_R G(x, y, z) \frac{|\nabla F|}{|\nabla F \cdot \hat{p}|} dA$$

4. **Explicit Form:** For $z = f(x, y)$ in xy plane:

$$\iint_S G(x, y, z) d\sigma = \iint_R \sqrt{f_x^2 + f_y^2 + 1} \, dxdy$$

1.13.3 Surface Integrals of Vector Fields:

1. Let $\vec{F}$ be a vector field in 3D space with continuous components defined over a smooth surface S having a chosen field of normal unit vector $\hat{n}$ orienting S , then the surface integral of $\vec{F}$ over S is:

$$SI = \iint_S (\vec{F} \cdot \hat{n}) d\sigma$$

2. This integral is also called the "flux" of the vector field $\vec{F}$.

3. **Parametric Form:** $\vec{r}(u, v) = f(u, v)\hat{i} + g(u, v)\hat{j} + h(u, v)\hat{k}$

$$\hat{n} = \frac{\vec{r}_u \times \vec{r}_v}{|\vec{r}_u \times \vec{r}_v|} \implies \iint_S (\vec{F} \cdot \hat{n}) d\sigma = \iint_R \vec{F} \cdot (\vec{r}_u \times \vec{r}_v) \, dudv$$

4. **Implicit Form:** $F(x, y, z) = C$

$$\hat{n} = \frac{\nabla F}{|\nabla F|} \implies \iint_S (\vec{F} \cdot \hat{n}) d\sigma = \iint_R \vec{F} \cdot \frac{\nabla F}{|\nabla F \cdot \hat{p}|} dA$$

1.13.4 Questions:

1. Find the surface area of the cone $z = \sqrt{x^2 + y^2}$ for $0 \leq r \leq 1$.

Solution: Let $x = r\cos\theta, y = r\sin\theta, z = r$

$$\implies \vec{r}(r, \theta) = r\cos\theta\hat{i} + r\sin\theta\hat{j} + r\hat{k}$$

$$\vec{r}_r \times \vec{r}_\theta = r\sin\theta\hat{i} - r\cos\theta\hat{j} - r\hat{k} \implies |\vec{r}_r \times \vec{r}_\theta| = \sqrt{2}r$$

$$A = \iint d\sigma = \int_0^{2\pi} \int_0^1 |\vec{r}_r \times \vec{r}_\theta| dr d\theta$$

$$\therefore A = 2\pi \int_0^1 (\sqrt{2}r) dr = \sqrt{2}\pi$$

2. Integrate $G(x, y, z) = xyz$ over the surface of the cube cut from the first octant while the lines $x = 1, y = 1, z = 1$.

Solution:

$$\text{Lower side } (z = 0) \implies G = xy(0) = 0 \implies SI_1 = 0$$

$$\text{Left side } (y = 0) \implies G = x(0)z = 0 \implies SI_2 = 0$$

$$\text{Back side } (x = 0) \implies G = (0)yz = 0 \implies SI_3 = 0$$

$$\text{Upper side } (z = 1) \implies G = xy(1) = xy, \quad \vec{F} = (z-1)\hat{k} \implies \nabla F = \hat{k}$$

$$\therefore SI_4 = \iint_R \vec{F} \cdot \frac{\nabla F}{|\nabla F \cdot \hat{p}|} dA = \int_0^1 \int_0^1 (xy) \frac{1}{1} dx dy = \frac{1}{4}$$

$$\text{Similarly, Right Side } (y = 1) \implies SI_5 = \frac{1}{4}$$

$$\text{and Front Side } (x = 1) \implies SI_6 = \frac{1}{4}$$

$$\therefore SI = \sum_{i=1}^6 SI_i = 0 + 0 + 0 + \frac{1}{4} + \frac{1}{4} + \frac{1}{4} = \frac{3}{4}$$

3. Evaluate $\iint_S \sqrt{x(1+2z)} d\sigma$ on the portion of the cylinder $y^2 = 2z$ over the triangular region $x \geq 0, y \geq 0, x + y \leq 1$ in the xy plane.

Solution:

$$z = f(x, y) = \frac{1}{2}y^2 \implies d\sigma = \sqrt{f_x^2 + f_y^2 + 1} dx dy = \sqrt{y^2 + 1} dx dy$$

$$SI = \iint_S \sqrt{x(1+2z)} d\sigma = \iint_R \sqrt{x(1+y^2)} \sqrt{y^2 + 1} dx dy$$

$$= \int_0^1 \int_0^{1-y} \sqrt{x(1+y^2)} dx dy = \frac{2}{3} \int_0^1 [x^{3/2}]_0^{1-y} (1+y^2) dy$$

$$= \frac{2}{3} \int_0^1 (1-y)^{3/2} (1+y^2) dy = \frac{284}{945}$$

4. Find the flux of the vector field $\vec{F} = yz\hat{i} + x\hat{j} - z^2\hat{k}$ through the cylinder $y = x^2$ ($0 \leq x \leq 1, 0 \leq z \leq 4$)

Solution: $\vec{r}(x, z) = x\hat{i} + x^2\hat{j} + z\hat{k} \implies \vec{r}_x = \hat{i} + 2x\hat{j}, \vec{r}_z = \hat{k}$

$$\vec{r}_x \times \vec{r}_z = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2x & 0 \\ 0 & 0 & 1 \end{vmatrix} = 2x\hat{i} - \hat{j}$$

$$\begin{aligned} \therefore SI &= \iint_S (\vec{F} \cdot \hat{n}) d\sigma = \iint_R \vec{F} \cdot \frac{\vec{r}_x \times \vec{r}_z}{|\vec{r}_x \times \vec{r}_z|} (|\vec{r}_x \times \vec{r}_z| dx dz) \\ &= \iint_R (x^2 z \hat{i} + x \hat{j} - z^2 \hat{k}) \cdot (2x \hat{i} - \hat{j}) dx dz = \int_0^1 \int_0^4 (2x^3 z - x) dx dz \\ &= \int_0^1 (16x^3 - 4x) dx = 2 \end{aligned}$$

1.13.5 Moment of Inertia and Center of Mass:

1. Mass, $M = \iiint_S \rho(x, y, z) d\sigma$

2. Centers of Mass:

$$\bar{x} = \frac{\iint_S \rho x d\sigma}{\iint_S \rho d\sigma} \quad \bar{y} = \frac{\iint_S \rho y d\sigma}{\iint_S \rho d\sigma} \quad \bar{z} = \frac{\iint_S \rho z d\sigma}{\iint_S \rho d\sigma}$$

3. Moments of Inertia:

$$\begin{aligned} I_x &= \iint_S (y^2 + z^2) \rho d\sigma & I_y &= \iint_S (x^2 + z^2) \rho d\sigma \\ I_z &= \iint_S (x^2 + y^2) \rho d\sigma & I_L &= \iint_S r^2(x, y, z) \rho d\sigma \end{aligned}$$

1.14 Stoke's Theorem:

Let S be a piecewise, smooth oriented surface having a piecewise smooth boundary curve C. Let $\vec{F} = M\hat{i} + N\hat{j} + P\hat{k}$ be a vector field whose components have continuous first partial derivatives on an open region containing S, then the circulation of $\vec{F}$ around C in the direction counterclockwise with respect to the surface unit normal vector $\hat{n}$ equals to the integration of curl of vector field $\vec{F}$ over S.

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_S (\vec{\nabla} \times \vec{F}) \cdot \hat{n} d\sigma$$

1.14.1 Questions:

1. Verify Stoke's theorem for the hemisphere $x^2 + y^2 + z^2 = 9$ ($z \geq 0$), it's bounding circle $x^2 + y^2 = 9$, $z \geq 0$ and the vector field $\vec{F} = y\hat{i} - x\hat{j}$

Solution: To prove: $\oint \vec{F} \cdot d\vec{r} = \iint_S (\vec{\nabla} \times \vec{F}) \cdot \hat{n} d\sigma$

LHS: Let $x = 3\cos\theta$, $y = 3\sin\theta$, $z = 0$

$$\Rightarrow \vec{r} = \langle 3\cos\theta, 3\sin\theta, 0 \rangle \Rightarrow \vec{v} = \langle -3\sin\theta, 3\cos\theta, 0 \rangle$$

$$\therefore \vec{F} = 3\sin\theta\hat{i} - 2\cos\theta\hat{j} \Rightarrow \vec{F} \cdot \vec{v} = -9\sin^2\theta - 9\cos^2\theta = -9$$

$$\Rightarrow \oint \vec{F} \cdot d\vec{r} = \int_0^{2\pi} (-9) d\theta = -18\pi$$

Now RHS: $x^2 + y^2 + z^2 - 9 = 0 \Rightarrow f(x, y, z) = x^2 + y^2 + z^2 - 9$

$$\Rightarrow \vec{\nabla} f = 2x\hat{i} + 2y\hat{j} + 2z\hat{k} \Rightarrow \vec{\nabla} f \cdot \hat{k} = 2z$$

$$\therefore \hat{n} d\sigma = \frac{\vec{\nabla} f}{|\vec{\nabla} f \cdot \hat{k}|} dxdy = \frac{2x\hat{i} + 2y\hat{j} + 2z\hat{k}}{2z} dxdy$$

$$\therefore \vec{\nabla} \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y & -x & 0 \end{vmatrix} = -2\hat{k}$$

$$\therefore \iint_S (\vec{\nabla} \times \vec{F}) \cdot \hat{n} d\sigma = \iint_R (-2\hat{k}) \cdot \left(\frac{2x\hat{i} + 2y\hat{j} + 2z\hat{k}}{2z} \right) dxdy$$

$$= \iint_R (-2) dxdy = -2 \cdot \text{Area}(R) = -2 \cdot 9\pi = -18\pi$$

Hence, verified Stoke's Theorem.

2. Verify Stoke's theorem for the vector field $\vec{F} = (2x - y)\hat{i} - yz^2\hat{j} - y^2z\hat{k}$

Solution: To prove: $\oint \vec{F} \cdot d\vec{r} = \iint_S (\vec{\nabla} \times \vec{F}) \cdot \hat{n} d\sigma$

LHS: Let $x = \cos\theta$, $y = \sin\theta$, $z = 0$

$$\Rightarrow \vec{r} = \langle \cos\theta, \sin\theta, 0 \rangle \Rightarrow \vec{v} = \langle -\sin\theta, \cos\theta, 0 \rangle$$

$$\Rightarrow \vec{F} \cdot \vec{v} = -(2\cos\theta - \sin\theta)(-\sin\theta) d\theta = (\sin^2\theta - \sin(2\theta)) d\theta$$

$$\Rightarrow \oint \vec{F} \cdot d\vec{r} = \int_0^{2\pi} (\sin^2\theta - \sin(2\theta)) d\theta = \pi$$

Now RHS: $x^2 + y^2 + z^2 - 1 = 0 \Rightarrow f(x, y, z) = x^2 + y^2 + z^2 - 1$

$$\Rightarrow \vec{\nabla} f = 2x\hat{i} + 2y\hat{j} + 2z\hat{k} \Rightarrow \vec{\nabla} f \cdot \hat{k} = 2z$$

$$\therefore \hat{n} d\sigma = \frac{\vec{\nabla} f}{|\vec{\nabla} f \cdot \hat{k}|} dxdy = \frac{2x\hat{i} + 2y\hat{j} + 2z\hat{k}}{2z} dxdy$$

$$\begin{aligned}
\therefore \vec{\nabla} \times \vec{F} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2x-y & -yz^2 & -y^2z \end{vmatrix} = \hat{k} \\
\therefore \iint_S (\vec{\nabla} \times \vec{F}) \cdot \hat{n} d\sigma &= \iint_R (\hat{k}) \cdot \left(\frac{2x\hat{i} + 2y\hat{j} + 2z\hat{k}}{2z} \right) dx dy \\
&= \iint_R (1) dx dy = \text{Area}(R) = \pi
\end{aligned}$$

Hence, verified Stoke's Theorem.

1.15 Divergence Theorem:

Let $\vec{F}$ be a vector field whose components have continuous first partial derivatives and let S be a piecewise smooth oriented closed surface. The flux of $\vec{F}$ across S in the direction of the surface outward unit normal field $\hat{N}$ equals to the triple integral of divergence of vector field over region D enclosed by the surface.

$$\iint_S \vec{F} \cdot \hat{n} d\sigma = \iiint_D (\vec{\nabla} \cdot \vec{F}) dV$$

Question: Find flux of $\vec{F} = x\hat{i} + y\hat{j} + z\hat{k}$ over the sphere $x^2 + y^2 + z^2 = a^2$.

Solution: $\vec{\nabla} \cdot \vec{F} = 1 + 1 + 1 = 3$

$$\therefore \phi = \iiint_D 3dV = 3(\text{Volume of Sphere}) = 4\pi a^3$$

2 Differential Equations

Differential Equation: An equation which involves the derivatives of one (or more) dependent variables wrt one (or more) independent variables.

2.1 Ordinary Differential Equation (ODE):

2.1.1 Some Definitions:

1. An ODE is a DE which involves the derivatives of one (or more) dependent variable wrt only one independent variable.
2. **Order of an ODE:** The order of an ODE is the highest order of the derivative in the equation.
3. **Degree of an ODE:** The degree of an ODE is the highest power of the highest order derivative in the equation.
4. **Linear ODE:** An ODE is called "linear" if all the dependent variable and their derivatives occur with degree one. Also no product of dependent variable and their derivative should occur.

2.1.2 Formation of an ODE:

Given function $f(x, y, c_1, c_2, \dots, c_n) = 0$, differentiate it n times.

Example: $y = a \cos x + b \sin x$

$$\begin{aligned} y' &= -a \sin x + b \cos x \implies y'' = -a \cos x - b \sin x = -(a \cos x + b \sin x) \\ &\implies y'' = -y \implies y + y'' = 0 \end{aligned}$$

2.2 Solution of an ODE with order n :

General Solution: A solution of given ODE is said to be a general solution if it contains as many arbitrary constants as the order of the ODE.

Particular Solution: A solution which is obtained from the general solution by assigning specific values to the arbitrary constants.

$y = f(x)$ is called the solution of an ODE of order n in an interval I if:

1. $y = f(x)$ is defined on I .
2. $y = f(x)$ has n derivatives on I .
3. $y = f(x)$ satisfies the given ODE.

2.2.1 Separation of Variables:

$$\frac{dy}{dx} = f(x)g(y) \implies \int \frac{dy}{g(y)} = \int f(x)dx$$

Example: $3e^x \tan(y)dx + (1 - e^x) \sec^2(y)dy = 0$

$$\begin{aligned} \frac{3e^x}{e^x - 1}dx &= \frac{\sec^2 y}{\tan y} dy \implies 3 \int \frac{d(e^x - 1)}{e^x - 1} = \int \frac{d(\tan y)}{\tan y} \\ \implies 3 \ln|e^x - 1| &= \ln|\tan y| + \ln|c| \implies (e^x - 1)^3 = c \tan y \end{aligned}$$

2.2.2 Equations Reducible to Separable Form:

For $y' = f(ax + by + c)$, let $ax + by + c = z$

$$\begin{aligned} \frac{dz}{dx} &= a + b \frac{dy}{dx} \implies \frac{dz}{dx} = a + by' \\ \implies \frac{dz}{dx} &= a + bf(z) \implies \int \frac{dz}{a + bf(z)} = \int dx \end{aligned}$$

Example: $y' = (4x + y + z)^2$

Let $z = 4x + y + z$.

$$\begin{aligned} \frac{dz}{dx} &= 4 + y' \implies \frac{dz}{dx} = 4 + z^2 \implies \int \frac{dz}{4 + z^2} = \int dx \\ \implies \frac{1}{2} \tan^{-1} \left(\frac{z}{2} \right) &= x + c \implies \frac{1}{2} \tan^{-1} \left(\frac{4x + y + 1}{2} \right) = x + c \end{aligned}$$

2.2.3 Homogeneous Differential Equation:

If we have a differential equation in terms of:

$$\frac{dy}{dx} = \phi \left(\frac{y}{x} \right)$$

Let

$$y = vx \implies \frac{dy}{dx} = v + x \frac{dv}{dx} = \phi(v)$$

Hence,

$$x \frac{dv}{dx} = \phi(v) - v \implies \int \frac{dv}{\phi(v) - v} = \int \frac{dx}{x}$$

Then substitute v back in terms of y and x .

Example: Solve: $xdy = y[\ln y - \ln x + 1]dx$

Let $y = vx \implies dy = vdx + xdv$

$$v + x \frac{dv}{dx} = v(\ln v + 1) = v \ln v + v$$

$$\frac{dv}{v \ln x} = \frac{dx}{x} \implies \frac{d(\ln v)}{\ln v} = \frac{dx}{x}$$

$$\ln(\ln v) = \ln(x) + \ln(c) \implies \ln v = cx \implies v = e^{cx}$$

Hence, solution:

$$y = xe^{cx} = xk^x$$

2.2.4 Equations Reducible to Homogeneous Form:

If we have a differential equation in terms of:

$$\frac{dy}{dx} = \frac{ax + by + c}{a'x + b'y + c'}$$

1. **If $a'x + b'y = k(ax + by)$:** Let $ax + by = z$, then

$$a + b \frac{dy}{dx} = \frac{dz}{dx} \implies \frac{dy}{dx} = \frac{1}{b} \left(\frac{dz}{dx} - a \right)$$

$$\frac{1}{b} \left(\frac{dz}{dx} - a \right) = \frac{z + c}{kz + c'} \implies \frac{dz}{dx} = b \left(\frac{z + c}{kz + c'} + \frac{a}{b} \right)$$

2. **For other cases:** Let $x = X + h$, $y = Y + k$. ($dX = dx$, $dY = dy$)

$$\frac{dY}{dX} = \frac{a(X + h) + b(Y + k) + c}{a'(X + h) + b'(Y + k) + c'} = \frac{aX + bY + (ah + bk + c)}{a'X + b'Y + (a'h + b'k + c')}$$

This becomes homogenous for $ah + bk + c = 0$ and $a'h + b'k + c' = 0$.
Solve for h and k .

$$\therefore \frac{dY}{dX} = \frac{aX + bY}{a'X + b'Y}$$

Now, solve homogenous differential equation by substituting $Y = vX$.
After solving, substitute back $Y = y - k$ and $X = x - h$.

2.2.5 Exact Differential Equations:

$Mdx + Ndy$ is said to be "exact" if there exists a $f(x, y)$ such that

$$Mdx + Ndy = d(f(x, y)) = 0$$

Theorem: Let $Mx + Ny = 0$ A necessary and sufficient condition for the above equation to be exact is: $M_y = N_x$.

And the solution of the equation is: $\int Mdx + \int (N - g_y)dy = c$.
where $g = \int Mdx$

Question: Solve: $y \sin(2x)dx - (1 - y^2 + \cos^2 x)dy = 0$ Here $M = y \sin(2x)$,
 $N = -(1 - y^2 + \cos^2 x)$

$$\therefore M_y = \sin(2x), N_x = -(-\sin(2x)) = \sin(2x)$$

As M_y and N_y are same, the equation is exact. Now,

$$\begin{aligned} g &= \int M dx = \int y \sin(2x) dx = -\frac{1}{2} y \cos(2x) \implies g_y = -\frac{1}{2} \cos(2x) \\ &-\frac{1}{2} y \cos(2x) + \int (y^2 - 1 - \cos^2 x + \frac{1}{2} \cos(2x)) dy = c \\ \therefore -\frac{1}{2} y \cos(2x) + \frac{1}{3} y^3 - y - y \cos^2 x + \frac{1}{2} y \cos(2x) &= c \end{aligned}$$

Hence,

$$\frac{1}{3} y^3 - y(1 + \cos^2 x) = c$$

2.3 Integrating Factor:

If $Mdx + Ndy = 0$ is not exact, then we can multiply some function $f(x, y)$ to make it exact. This function is called the integrating factor.

1. **If $Mdx + Ndy = 0$ is not exact, and M, N are homogenous functions of same degree:** IF = $\frac{1}{Mx+Ny}$

Question: Solve: $(3xy^2 - y^3)dx - (2x^2y - xy^2)dy = 0$

Solution:

$$\begin{aligned} M &= 3xy^2 - y^3 \implies M_y = 6xy - 3y^2 \\ N &= 2x^2y - xy^2 \implies N_x = 4xy - y^2 \end{aligned}$$

As $M_y \neq N_x \implies$, equation is not exact. Also, M and N are homogenous functions of same degree.

$$\therefore IF = \frac{1}{Mx + Ny} = \frac{1}{(3xy^2 - y^3)x + (2x^2y - xy^2)y} = \frac{1}{x^2y^2}$$

Now,

$$\begin{aligned} \frac{3xy^2 - y^3}{x^2y^2} dx + \frac{2x^2y - xy^2}{x^2y^2} dy &= 0 \\ \therefore M &= \frac{3}{x} - \frac{y}{x^2} \implies M_y = -\frac{1}{x^2} \\ \therefore N &= -\frac{2}{y} + \frac{1}{x} \implies N_x = -\frac{1}{x^2} \end{aligned}$$

Hence, $M_y = N_x \implies$ the equation is exact.

$$\begin{aligned} g &= \int M dx = \int \left(\frac{3}{x} - \frac{y}{x^2} \right) dx = 3 \ln|x| + \frac{y}{x} \implies g_y = \frac{1}{x} \\ \therefore 3 \ln|x| + \frac{y}{x} + \int \left(-\frac{2}{y} + \frac{1}{x} \right) dy &= c \\ \therefore 3 \ln|x| + \frac{y}{x} - 2 \ln|y| + \ln|x| &= c \implies 3 \ln|x| - 2 \ln|y| + \frac{y}{x} = c \end{aligned}$$

Hence,

$$\ln \left| \frac{x^3}{y^2} \right| + \frac{y}{x} = c$$

2. If $Mdx + Ndy = 0$ is not exact, and if $Mdx + Ndy = yf_1(x, y)dx + xf_2(x, y)dy = 0$: IF = $\frac{1}{Mx - Ny}$

Question: Solve: $(y + xy^2)dx + (x - x^2y)dy = 0$

Solution:

$$M = y + xy^2 \implies M_y = 1 + 2xy$$

$$N = x - x^2y \implies N_x = 1 - 2xy$$

As $M_y \neq N_x \implies$, equation is not exact. Also, the given equation can be re-written as: $y(1 + xy)dx + x(1 - xy)dy = 0$

$$\therefore IF = \frac{1}{Mx - Ny} = \frac{1}{x(y + xy^2) - y(x - x^2y)} = \frac{1}{2x^2y^2}$$

Now,

$$\frac{y + xy^2}{2x^2y^2}dx + \frac{x - x^2y}{2x^2y^2}dy = 0$$

$$\therefore M = \frac{1}{2x^2y} + \frac{1}{2x} \implies M_y = -\frac{1}{2x^2y^2}$$

$$\therefore N = \frac{1}{2xy^2} - \frac{1}{2y} \implies N_x = -\frac{1}{2x^2y^2}$$

Hence, $M_y = N_x \implies$ the equation is exact.

$$g = \int Mdx = \int \left(\frac{1}{2x^2y} + \frac{1}{2x} \right) dx = \frac{-1}{2xy} + 2\ln|x| \implies g_y = \frac{1}{2xy^2}$$

$$\therefore \frac{-1}{2xy} + 2\ln|x| + \int \left(\frac{1}{2xy^2} - \frac{1}{2y} - \frac{1}{2xy^2} \right) dy = c$$

$$\therefore \frac{-1}{2xy} + 2\ln|x| - 2\ln|y| = c \implies \frac{-1}{2xy} + 2\ln|x| - 2\ln|y| = c$$

3. If $Mdx + Ndy = 0$ is not exact, and if $\frac{M_y - N_x}{N} = \psi(x)$: IF = $e^{\int \psi(x)dx}$

Question: Solve $(x^2 + y^2 + x)dx + xydy = 0$

Solution:

$$M = x^2 + y^2 + x \implies M_y = 2y$$

$$N = xy \implies N_x = y$$

As $M_y \neq N_x \implies$, equation is not exact. Now,

$$\frac{M_y - N_x}{N} = \frac{2y - y}{xy} = \frac{1}{x} = \psi(x)$$

$$\therefore IF = e^{\int \psi(x)dx} = e^{\ln|x|} = x$$

Now,

$$x(x^2 + y^2 + x)dx + x(xy)dy = 0 \implies (x^3 + xy^2 + x^2)dx + x^2ydy = 0$$

$$\therefore M = x^3 + xy^2 + x^2 \implies M_y = 2xy$$

$$N = x^2y \implies N_x = 2xy$$

As $M_y = N_x \implies$ the equation is exact.

$$g = \int Mdx = \int (x^3 + xy^2 + x^2)dx = \frac{1}{4}x^4 + \frac{1}{2}x^2y^2 + \frac{1}{3}x^3 \implies g_y = x^2y$$

$$\therefore \frac{1}{4}x^4 + \frac{1}{2}x^2y^2 + \frac{1}{3}x^3 + \int (x^2y - x^2y)dy = c$$

$$\therefore \frac{1}{4}x^4 + \frac{1}{2}x^2y^2 + \frac{1}{3}x^3 = c$$

4. If $Mdx + Ndy = 0$ is not exact, and if $\frac{N_x - M_y}{M} = \phi(y)$: IF $= e^{\int \phi(y)dy}$
Question: Solve $(2xy^4e^y + 2xy^3 + y)dx + (x^2y^4e^y - x^2y^2 - 3x)dy = 0$
Solution:

$$M = 2xy^4e^y + 2xy^3 + y \implies M_y = 2xy^4e^y + 8xy^3e^y + 6xy^2 + 1$$

$$N = x^2y^4e^y - x^2y^2 - 3x \implies N_x = 2xy^4e^y - 2xy^2 - 3$$

As $M_y \neq N_x \implies$, equation is not exact. Now,

$$\begin{aligned} \frac{N_x - M_y}{M} &= \frac{2xy^4e^y - 2xy^2 - 3 - (2xy^4e^y + 8xy^3e^y + 6xy^2 + 1)}{2xy^4e^y + 2xy^3 + y} \\ &= \frac{-2xy^2 - 3 - 8xy^3e^y - 6xy^2 - 1}{2xy^4e^y + 2xy^3 + y} = \frac{-4(xy^4e^y + 2xy^3 + y)}{y(xy^4e^y + 2xy^3 + y)} = \frac{-4}{y} = \phi(y) \\ \therefore IF &= e^{\int \phi(y)dy} = e^{-4\ln|y|} = \frac{1}{y^4} \end{aligned}$$

Now,

$$\frac{2xy^4e^y + 2xy^3 + y}{y^4}dx + \frac{x^2y^4e^y - x^2y^2 - 3x}{y^4}dy = 0$$

$$\therefore M = 2xe^y + 2xy^{-1} + y^{-3} \implies M_y = 2xe^y - 2xy^{-2} - 3y^{-4}$$

$$\therefore N = x^2e^y - x^2y^{-2} - 3xy^{-4} \implies N_x = 2xe^y - 2xy^{-2} - 3y^{-4}$$

As $M_y = N_x \implies$ the equation is exact.

$$g = \int Mdx = \int (2xe^y + 2xy^{-1} + y^{-3})dx = x^2e^y + x^2y^{-1} + xy^{-3}$$

$$\implies g_y = x^2e^y - x^2y^{-2} - 3xy^{-4}$$

$$\therefore x^2e^y + x^2y^{-1} + xy^{-3} + \int (x^2e^y - x^2y^{-2} - 3xy^{-4} - x^2e^y + x^2y^{-2} + 3xy^{-4})dy = c$$

$$\therefore x^2e^y + x^2y^{-1} + xy^{-3} = c$$

5. If $Mdx + Ndy = 0$ is not exact, and if it can be expressed as

$$x^a y^b [mydx + nxdy] + x^{a'} + y^{b'} [m'ydx + n'xdy] = 0$$

where $a, a', b, b', m, m', n, n'$ are constants then: IF = $x^h y^k$ where h and k can be obtained from:

$$\frac{a+h+1}{m} = \frac{b+k+1}{n} \text{ and } \frac{a'+h+1}{m'} = \frac{b'+k+1}{n'}$$

Question: Solve: $(y^2 + 2x^2y)dx + (2x^3 - xy)dy = 0$

Solution:

$$M = y^2 + 2x^2y \implies M_y = 2y + 2x^2$$

$$N = 2x^3 - xy \implies N_x = 6x^2 - y$$

As $M_y \neq N_x \implies$, equation is not exact. Now, the above can be re-written as:

$$y(ydx - xdy) + x^2(2ydx + 2xdy) = 0$$

Hence, $a = 0, b = 1, m = 1, n = -1, a' = 2, b' = 0, m' = 2, n' = 2$. Now,

$$\frac{0+h+1}{1} = \frac{1+k+1}{-1} \implies h = -k - 3$$

$$\frac{2+h+1}{2} = \frac{0+k+1}{2} \implies h = k - 2$$

Hence, on solving $k = -1/2$, and $h = -5/2$.

$$\therefore IF = x^{-5/2} y^{-1/2}$$

Now,

$$x^{-5/2} y^{-1/2} (y^2 + 2x^2y)dx + x^{-5/2} y^{-1/2} (2x^3 - xy)dy = 0$$

$$(x^{-5/2} y^{3/2} + 2x^{-1/2} y^{1/2})dx + (2x^{1/2} y^{-1/2} - x^{-3/2} y^{1/2})dy = 0$$

$$\therefore M = x^{-5/2} y^{3/2} + 2x^{-1/2} y^{1/2} \implies M_y = \frac{3}{2} x^{-5/2} y^{1/2} + x^{-1/2} y^{-1/2}$$

$$\therefore N = 2x^{1/2} y^{-1/2} - x^{-3/2} y^{1/2} \implies N_x = \frac{3}{2} x^{-5/2} y^{1/2} + x^{-1/2} y^{-1/2}$$

As $M_y = N_x \implies$ the equation is exact.

$$g = \int Mdx = \int (x^{-5/2} y^{3/2} + 2x^{-1/2} y^{1/2})dx = -\frac{2}{3} x^{-3/2} y^{3/2} + 4x^{1/2} y^{1/2}$$

$$\implies g_y = -x^{-3/2} y^{1/2} + 2x^{1/2} y^{-1/2}$$

$$\therefore -\frac{2}{3} x^{-3/2} y^{3/2} + 4x^{1/2} y^{1/2} + \int (-x^{-3/2} y^{1/2} - 2x^{1/2} y^{-1/2} - x^{-3/2} y^{1/2} + 2x^{1/2} y^{-1/2})dy = c$$

$$\therefore -\frac{2}{3} x^{-3/2} y^{3/2} + 4x^{1/2} y^{1/2} = c$$

2.3.1 First Order Linear Differential Equations:

1. **General Form:** $y' + p(x)y = q(x)$
2. **Integrating Factor:** $IF = e^{\int p(x)dx}$
3. **Solution:** $y(IF) = \int (IF \cdot q(x))dx$
4. **Question:** Solve: $y' \cos x + y \sin x = 1$

Solution:

$$y' \cos x + y \sin x = 1 \implies y' + \tan x y = \sec x$$

$$IF = e^{\int \tan x dx} = e^{\ln|\sec x|} = \sec x$$

$$y(\sec x) = \int (\sec x \cdot \sec x dx) = \int \sec^2 x dx = \tan x + c$$

2.3.2 Bernoulli's Equation:

1.

$$\frac{dy}{dx} + y \cdot p(x) = y^n \cdot q(x) \implies \frac{1}{y^n} \frac{dy}{dx} + \frac{1}{y^{n-1}} p(x) = q(x)$$

$$\text{Let } u = y^{1-n} \implies \frac{du}{dx} = (1-n) \frac{1}{y^n} \frac{dy}{dx}$$

$$\implies \frac{-1}{n-1} \frac{du}{dx} + u \cdot p(x) = q(x)$$

2. **Question:** Solve: $y' + \frac{y}{x} = y^2$

Solution:

$$\frac{dy}{dx} + \frac{y}{x} = y^2 \implies \frac{1}{y^2} \frac{dy}{dx} + \frac{1}{xy} = 1$$

$$\text{Let } u = \frac{1}{y} \implies \frac{du}{dx} = -\frac{1}{y^2} \frac{dy}{dx} \implies \frac{du}{dx} - \frac{1}{x} = -1$$

$$\text{Hence, IF} = e^{\int -(1/x)dx} = \frac{1}{x}$$

$$u \left(\frac{1}{x} \right) = \int - \left(\frac{1}{x} \cdot 1 \right) dx = -\ln|x| + c \implies \frac{1}{xy} = -\ln|x| + c$$

2.3.3 Equations Reducible to LDE:

Let $f'(y)y' + f(y) \cdot p(x) = q(x)$

Now, let $f(y) = u \implies f'(y)y' = u'$

$\implies u' + up(x) = q(x)$

Question:

$$\text{Solve: } \frac{dy}{dx} + \frac{y}{x} \ln(y) = \frac{y \ln^2(y)}{x^2}$$

Solution:

$$\frac{dy}{dx} + \frac{y}{x} \ln(y) = \frac{y \ln^2(y)}{x^2} \implies \frac{1}{y \ln^2(y)} \frac{dy}{dx} + \frac{1}{x \ln(y)} = \frac{1}{x^2}$$

$$\text{Let } u = \frac{1}{\ln(y)} \implies \frac{du}{dx} = -\frac{1}{y \ln^2(y)} \frac{dy}{dx}$$

$$\implies \frac{du}{dx} - \frac{1}{x} u = \frac{-1}{x^2} \implies IF = \frac{1}{x}$$

$$u \left(\frac{1}{x} \right) = \int - \left(\frac{1}{x} \cdot \frac{-1}{x^2} \right) dx = -\frac{1}{2x^2} + c$$

$$\text{Hence, } \frac{1}{y} = \frac{1}{2x} + cx$$

2.4 LDE of Higher Orders with Constant Coefficients:

1. Consider:

$$a_0 \frac{d^n y}{dx^n} + a_1 \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_n y = f(x)$$

$$\implies a_0 D^n + a_1 D^{n-1} + \dots + a_n y = f(x)$$

where $a_0, a_1, \dots, a_n$ are constants and $D = d/dx$

2. General Solution is: $y = y_c + y_p$

where y_c = complementary function: $f(D)y_c = 0$

and y_p = particular integral: $f(D)y_p = f(x)$

2.4.1 Finding the Complementary Function:

For y_c , $(D - \lambda_1)(D - \lambda_2) \dots (D - \lambda_n)y = 0$

$$(D - \lambda_1)y = 0 \implies \frac{dy}{dx} - \lambda_1 y \implies y = c_1 e^{\lambda_1 x}$$

Now, auxiliary equation, $a_0 m^n + a_1 m^{n-1} + \dots + a_n = 0$

1. When roots are real and distinct $(\lambda_1, \lambda_2, \dots, \lambda_n)$:

Solution: $y = c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x} + \dots + c_n e^{\lambda_n x}$

2. When roots are real and equal $(\lambda_1, \lambda_1, \lambda_3, \dots, \lambda_n)$:

Solution: $y = (c_1 + c_2 x) e^{\lambda_1 x} + c_3 e^{\lambda_3 x} + \dots + c_n e^{\lambda_n x}$

3. When roots are imaginary $(\alpha \pm i\beta, \lambda_3, \dots, \lambda_n)$:

Solution: $y = e^{\alpha x} [c_1 \cos(\beta x) + c_2 \sin(\beta x)] + c_3 e^{\lambda_3 x} + \dots + c_n e^{\lambda_n x}$

4. When roots are imaginary and repeated $(\alpha \pm i\beta, \alpha \pm i\beta, \lambda_5, \dots, \lambda_n)$:

Solution:

$y = e^{\alpha x} [(c_1 + c_2 x) \cos(\beta x) + (c_3 + c_4 x) \sin(\beta x)] + c_5 e^{\lambda_5 x} + \dots + c_n e^{\lambda_n x}$

5. When roots are irrational $(\alpha \pm \sqrt{\beta}, \lambda_3, \dots, \lambda_n)$:

Solution: $y = e^{\alpha x} [c_1 \cosh(\sqrt{\beta} x) + c_2 \sinh(\sqrt{\beta} x)] + c_3 e^{\lambda_3 x} + \dots + c_n e^{\lambda_n x}$

2.4.2 Finding the Particular Integral

For y_p , $(D - \alpha)y = f(x) \implies y_p = \frac{f(x)}{D - \alpha}$

$$\frac{dy}{dx} - \alpha y = f(x) \implies IF = e^{\int -\alpha dx} = e^{-\alpha x}$$

$$\therefore \text{Solution: } ye^{-\alpha x} = \int e^{-\alpha x} f(x) dx + c$$

$$\therefore y = e^{\alpha x} \int e^{-\alpha x} f(x) dx + e^{\alpha x} c$$

But $ce^{\alpha x} = y_c \implies y_p = e^{\alpha x} \int e^{-\alpha x} f(x) dx$ Hence,

$$\frac{f(x)}{D - \alpha} = e^{\alpha x} \int e^{-\alpha x} f(x) dx$$

1. **Question:** Find y_p : $(D^2 - 3D + 2)y = \cos(e^{-x})$

Solution:

$$(D - 1)(D - 2)y_p = \cos(e^{-x}) \implies y_p = \frac{1}{D - 2} \left[\frac{1}{D - 1} \cos(e^{-x}) \right]$$

$$\therefore y_p = \frac{1}{D - 2} \left[e^x \int e^{-x} \cos(e^{-x}) dx \right] = \frac{1}{D - 2} [-e^x \sin(e^{-x})]$$

$$\implies y_p = -e^{2x} \int e^{-2x} e^x \sin(e^{-x}) dx = -e^{2x} \int e^{-x} \sin(e^{-x}) dx$$

$$\implies y_p = -e^{2x} \cos(e^{-x})$$

Special Case 1: If $f(x) = e^{ax} \implies L(D)y = e^{ax} \implies y = \frac{1}{L(D)} e^{ax}$

then replace D with a $\implies y = \frac{1}{L(a)} e^{ax}$

In case of failure (i.e. $L(a) = 0$), differentiate the denominator wrt D and multiply x to the numerator.

1. **Question:** Find y_p : $(D^2 - 5D + 6)y = \cosh(x)$

Solution:

$$y_p = \frac{1}{D^2 - 5D + 6} \left(\frac{e^x + e^{-x}}{2} \right) = \frac{1}{2} \left[\frac{1}{D^2 - 5D + 6} e^x + \frac{1}{D^2 - 5D + 6} e^{-x} \right]$$

Replacing D with 1, we get:

$$y_p = \frac{1}{2} \left[\frac{1}{(1)^2 - 5(1) + 6} e^x + \frac{1}{(1)^2 - 5(1) + 6} e^{-x} \right] = \frac{6e^x + e^{-x}}{24}$$

2. **Question:** Find y_p : $(4D^2 + 12D + 9)y = 144e^{-3x}$

Solution:

$$y_p = \frac{144}{4D^2 + 12D + 9} e^{-3x} = \frac{144}{(2D + 3)^2} e^{-3x}$$

Replacing D with -3, we get:

$$y_p = \frac{144}{(2(-3) + 3)^2} e^{-3x} = \frac{144}{9} e^{-3x} = 16e^{-3x}$$

3. **Question:** Find y_p : $(D^2 + 4D + 4)y = e^{-2x}$

Solution:

$$y_p = \frac{1}{D^2 + 4D + 4} e^{-2x} = \frac{1}{(D + 2)^2} e^{-2x}$$

If we replace D with -2, we get 1/0 form, hence differentiate the denominator wrt D and multiply x to the numerator.

$$y_p = \frac{1}{2D + 4} x e^{-2x} \frac{1}{2} x^2 e^{-2x}$$

Special Case 2: If $f(x) = \sin(ax) / \cos(ax)$, then replace D^2 with $-a^2$

1. **Question:** Find y_p : $(D^2 - 3D + 2)y = \sin(3x)$

Solution:

$$y_p = \frac{1}{D^2 - 3D + 2} \sin(3x) = \frac{1}{(-9) - 3D + 2} \sin(3x) = -\frac{1}{3D + 7} \sin(3x)$$

$$y_p = -\frac{3D - 7}{9D^2 - 49} \sin(3x) = -\frac{3D - 7}{9(-9) - 49} \sin(3x) = \frac{7 - 3D}{-130} \sin(3x)$$

$$y_p = \frac{3}{130} \frac{d}{dx} \sin(3x) - \frac{7}{130} \sin(3x) \implies y_p = \frac{9\cos(3x) - 7\sin(3x)}{130}$$

2. **Question:** Find y_p : $(D^2 + 4)y = \sin(2x)$

Solution: If we replace D^2 with -4, we get 1/0 form. Hence, differentiate the denominator wrt D and multiply x to the numerator.

$$y_p = \frac{1}{D^2 + 4} \sin(2x) = \frac{x}{2D} \sin(2x) = \frac{x}{2} \int \sin(2x) dx = -\frac{x}{4} \cos(2x)$$

3. **Question:** Find y_p : $(D^2 + 4D + 4)y = \sin^2 x$

Solution:

$$y_p = \frac{1}{D^2 + 4D + 4} \sin^2 x = \frac{1}{D^2 + 4D + 4} \left(\frac{1 - \cos(2x)}{2} \right)$$

$$y_p = \frac{1}{2} \left[\frac{1}{D^2 + 4D + 4} e^{0x} - \frac{1}{D^2 + 4D + 4} \cos(2x) \right]$$

$$y_p = \frac{1}{2} \left[\frac{1}{0 + 0 + 4} - \frac{1}{-4 + 4D + 4} \cos(2x) \right] = \frac{1}{2} \left(\frac{1}{4} - \frac{1}{4} \int \cos(2x) dx \right)$$

$$y_p = \frac{1}{16} (2 - \sin(2x))$$

4. **Question:** Find y_p : $(D^3 + 1)y = \cos(2x + 1)$

Solution:

$$y_p = \frac{1}{D^3 + 1} \cos(2x + 1) = \frac{1}{D(-4) + 1} \cos(2x + 1) = \frac{1 + 4D}{1 - 16D^2} \cos(2x + 1)$$

$$y_p = \frac{1+4D}{1-16(-4)} \cos(2x+1) = \frac{1}{65} [\cos(2x+1) + 4D(\cos(2x+1))]$$

$$\therefore y_p = \frac{1}{65} [\cos(2x+1) - 8\sin(2x+1)]$$

Special Case 3: If $f(x)$ is a polynomial in x , then the least power of $f(D)$ should be taken common.

$$y_p = \frac{1}{f(D)} f(x) = \frac{1}{D^r(1-\psi(D))} f(x) = \frac{1}{D^r} (1-\psi(D))^{-1} f(x)$$

Then apply the expansion $(1-x)^{-1} = 1+x+x^2+\dots$ to get the solution.

1. **Question:** Find y_p : $(D^2 + D)y = x^2 + 2x + 4$

Solution:

$$y_p = \frac{1}{D^2 + D} (x^2 + 2x + 4) = \frac{1}{D(D+1)} (x^2 + 2x + 4)$$

$$y_p = \frac{1}{D} (1+D)^{-1} (x^2 + 2x + 4) = \frac{1}{D} (1-D+D^2) (x^2 + 2x + 4)$$

$$= \int [(x^2 + 2x + 4) - (2x + 2) + (2)] dx = \int (x^2 + 4) dx = \frac{1}{3} x^3 + 4x$$

2. **Question:** Find y_p : $(D^4 - a^4)y = x^4$

Solution:

$$y_p = \frac{1}{D^4 - a^4} x^4 = -\frac{1}{a^4(1 - \frac{D^4}{a^4})} x^4 = -\frac{1}{a^4} \left(1 - \frac{D^4}{a^4}\right)^{-1} x^4$$

$$y_p = \frac{1}{a^4} \left(1 + \frac{D^4}{a^4}\right) x^4 = \frac{1}{a^8} (a^4 x^4 - D^4(x^4))$$

$$\therefore y_p = \frac{1}{a^8} (a^4 x^4 - 24)$$

Special Case 4:

$$\text{If } f(x) = e^{ax} v(x) \implies y_p = \frac{1}{f(D)} e^{ax} v(x) = e^{ax} \frac{1}{f(D+a)} v(x)$$

1. **Question:** Find y_p : $(D^2 - 2D + 4)y = e^x \cos x$

Solution:

$$y_p = \frac{1}{D^2 - 2D + 4} e^x \cos x = e^x \frac{1}{(D+1)^2 - 2(D+1) + 4} \cos x$$

$$y_p = e^x \frac{1}{D^2 + 3} \cos x = e^x \frac{1}{-1 + 3} \cos x = \frac{1}{2} e^x \cos x$$

Special Case 5:

$$\text{If } f(x) = xv(x) \implies y_p = x \frac{1}{\phi(D)} v(x) + \frac{d}{dD} \left(\frac{1}{\phi(D)} \right) v(x)$$

1. **Question:** Find y_p : $(D^2 + 9)y = x \sin x$

Solution:

$$y_p = \frac{1}{D^2 + 9} \sin x = x \frac{1}{D^2 + 9} \sin x + \frac{-2D}{(D^2 + 9)^2} \sin x = \frac{4 \sin x - \cos x}{32}$$

2.5 Cauchy-Euler Equations:

Consider $(a_0 x^n D^n + a_1 x^{n-1} D^{n-1} + \dots + a_n)y = f(x)$

Let $x = e^z \implies z = \ln(x) \implies \frac{dy}{dz} = x \frac{dy}{dx}$

Hence, $xDy = \theta y \implies xD = \theta, \quad x^2 D^2 = \theta(\theta - 1), \dots$

1. **Question:** Solve: $(x^2 D^2 + 1)y = x^2$

Solution: Let $x = e^z \implies z = \ln(x)$

$$(\theta(\theta - 1) + 1)y = e^{2z} \implies (\theta^2 - \theta + 1)y = e^{2z}$$

$$\text{For } y_c, m^2 - m + 1 = 0 \implies m = \frac{1}{2}(1 \pm \sqrt{3}i)$$

$$\therefore y_c = e^{\frac{1}{2}z} \left(c_1 \cos \left(\frac{\sqrt{3}}{2} z \right) + c_2 \sin \left(\frac{\sqrt{3}}{2} z \right) \right)$$

For y_p :

$$y_p = \frac{1}{\theta^2 - \theta + 1} e^{2z} = \frac{1}{4 - 2 + 1} e^{2z} = \frac{1}{3} e^{2z}$$

Hence,

$$y = \frac{1}{3} x^2 + \sqrt{x} \left(c_1 \cos \left(\frac{\sqrt{3}}{2} \ln(x) \right) + c_2 \sin \left(\frac{\sqrt{3}}{2} \ln(x) \right) \right)$$

2. **Question:** Solve: $(x^2 D^2 - 3xD + 4)y = 2x^2$

Solution: Let $x = e^z \implies z = \ln(x)$

$$(\theta(\theta - 1) - 3\theta + 4)y = 2e^{2z} \implies (\theta - 2)^2 y = 2e^{2z}$$

$$\text{For } y_c: m^2 - 4m + 4 = 0 \implies m = 2$$

$$y_c = (c_1 + c_2 x) e^{2z}$$

For y_p :

$$y_p = \frac{2}{(\theta - 2)^2} e^{2z} = \frac{z}{2(\theta - 1)} e^{2z} = \frac{z^2}{2} e^{2z} = z^2 e^{2z}$$

Hence,

$$y = x^2 \ln^2(x) + x^2 (c_1 + c_2 \ln(x))$$

2.6 Method of Undetermined Coefficients:

1. Limitations:

- (a) Only y_p can be found.
- (b) LDE of constant coefficients.
- (c) RHS should be a combination of UC Functions.

2. UC Functions: $x^n, e^x, \sin(ax + b)$ and their products.

3. UC Set: The set of all differentials of a given functions without their coefficients.

4. Procedure:

- (a) Find UC Set of each expression of RHS (say, $S_1, S_2, \dots$)
Example: $(D^2 + 1)y = \sin x + x \sin x$
 $S_1 = \{\sin x, \cos x\}, S_2 = \{x \sin x, x \cos x, \sin x, \cos x\}$
- (b) Find y_c for the given LDE.
Example: $m^2 + 1 = 0 \implies y_c = c_1 \cos(x) + c_2 \sin(x)$
- (c) If any S_i is contained in another UC Set S_j , then remove S_i .
Example: $S_2 = \{x \sin x, x \cos x, \sin x, \cos x\}$
- (d) If y_c is contained in any S_j , then multiply the least positive integral power of x to each factor of S_j .
Example: $S_2 = \{x^2 \sin x, x^2 \cos x, x \sin x, x \cos x\}$
- (e) Write out the set in a linear combination.
Example: $y_p = Ax^2 \sin x + Bx^2 \cos x + Cx \sin x + Dx \cos x$

5. Question: Find trial solution: $(D^2 - 3D + 2)y = xe^x \sin x$

Solution: UC Set: $S = \{xe^x \sin x, x^2 \sin x, x^2 \cos x, x \sin x, x \cos x\}$

$$\text{For } y_c, m^2 - 3m + 2 = 0 \implies m = 1, 2 \implies y_c = c_1 e^x + c_2 e^{2x}$$

As there is only one UC, no removal can be done. Similarly, as y_c is not contained in S , no multiplication is required.

$$\therefore y_p = Ax^2 \sin x + Bx^2 \cos x + Cx \sin x + Dx \cos x$$

2.7 Method of Variation of Parameters:

1. Consider a second order differential equation:

$$y'' + P(x)y' + Q(x)y = R(x)$$

$$\text{Let } y_c = c_1 u(x) + c_2 v(x) \implies \boxed{y_p = A(x)u(x) + B(x)v(x)}$$

$$\text{where, } A(x) = - \int \frac{vR}{w} dx, \quad B(x) = \int \frac{uR}{w} dx, \quad w = \begin{vmatrix} u & v \\ u' & v' \end{vmatrix}$$

2. **Question:** Solve: $(D^2 + 1)y = \sec^2 x$

Solution: $y_c = c_1 \cos x + c_2 \sin x \implies u = \cos x, y = \sin x$

$$w = \begin{vmatrix} u & v \\ u' & v' \end{vmatrix} = \begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix} = \cos^2 x + \sin^2 x = 1$$

$$A(x) = - \int \frac{\sin x \cdot \sec^2 x}{1} dx = - \int \sec x \tan x dx = -\sec x$$

$$B(x) = \int \frac{\cos x \cdot \sec^2 x}{1} dx = \int \sec x dx = \ln |\sec x + \tan x|$$

Hence,

$$y_p = -\sec x \cdot \cos x + \ln |\sec x + \tan x| \cdot \sin x = \sin x \cdot \ln |\sec x + \tan x| - 1$$

2.7.1 Wronskian:

Let $\phi_1, \phi_2, \dots, \phi_n$ have $n-1$ derivatives, then:

$$W(\phi_1, \phi_2, \dots, \phi_n) = \begin{vmatrix} \phi_1 & \phi_2 & \dots & \phi_n \\ \phi_1' & \phi_2' & \dots & \phi_n' \\ \vdots & \vdots & \ddots & \vdots \\ \phi_1^{(n-1)} & \phi_2^{(n-1)} & \dots & \phi_n^{(n-1)} \end{vmatrix}$$

$\phi_1, \phi_2, \dots, \phi_n$ are linearly independent iff $W \neq 0$.

$\phi_1, \phi_2, \dots, \phi_n$ are linearly dependent iff $W = 0$.

2.7.2 Abel's Formula:

For $a_0(x)y'' + a_1(x)y' + a_2(x)y + a_3 = 0$

where $a_0 \neq 0$ and a_0, a_1, a_2, a_3 are continuous. Let y_1 and y_2 be two solutions, then:

$$\text{Abel's Formula: } W = ce^{\int \frac{a_1(x)}{a_0(x)} dx}$$

2.7.3 Lipschitz Condition:

A function $f(x, y)$ is set to satisfy Lipschitz condition if there exists a positive real number k such that $|f(x, y_1) - f(x, y_2)| \leq k \forall (x, y_1), (x, y_2) \in \mathbb{R}$ or $\frac{\partial f}{\partial y} \leq k$.

2.8 Picard's Theorem for Existence of Uniqueness of a solution of a IVP.

Let $y' = f(x, y)$ and $y(x_0) = y_0$

Consider a region R : $|x - x_0| < a, \quad |y - y_0| < b$.

IF

1. $f(x, y)$ is continuous in R .

2. $f(x, y)$ is bounded and there exists a positive real number M such that $|f(x, y)| \leq M \forall g(x, y) \in R$
3. $f(x, y)$ satisfies L.C.

THEN there exists a unique solution in $|x - x_0| < h$ where $h = \min\{a, \frac{b}{M}\}$

3 Laplace Transforms

3.1 Introduction:

Let $f(t)$ be a function of variable t , defined for all positive values of t . Let s be a positive real constant, then Laplace transform of f is:

$$L(f(t)) = \int_0^{\infty} e^{-st} f(t) dt$$

3.1.1 Piecewise Continuous Function:

A function $f(t)$ is said to be "piecewise continuous" in any interval $[a, b]$ if it is defined on the interval and that interval can be divided to a finite number of subintervals in each of which $f(t)$ is continuous.

3.1.2 Exponential Order:

A function $f(t)$ is said to be of exponential order if:

$$\lim_{t \rightarrow \infty} [e^{-st} f(t)] = 0$$

1. **Question:** Check if $f(t) = t^2$ is of exponential order.

Solution:

$$\lim_{t \rightarrow \infty} [e^{-st} t^2] = \lim_{t \rightarrow \infty} \left[\frac{t^2}{e^{st}} \right] = 0$$

Hence, $f(t) = t^2$ is of exponential order.

2. **Question:** Check if $f(t) = e^{t^2}$ is of exponential order.

Solution:

$$\lim_{t \rightarrow \infty} [e^{-st} e^{t^2}] = \lim_{t \rightarrow \infty} [e^{t^2 - st}] = \infty$$

Hence, $f(t) = e^{t^2}$ is not of exponential order.

3.1.3 Sufficient condition for existence of Laplace Transform:

Let $f(t)$ be defined and continuous for all positive values of t . Then the Laplace transform of $f(t)$ exists if the following conditions are satisfied:

1. $f(t)$ is piecewise continuous.

2. $f(t)$ is of exponential order.

Question: Find Laplace Transforms of the following functions:

1. $f(t) = e^{at}$

Solution:

$$L(e^{at}) = \int_0^{\infty} e^{-st} e^{at} dt = \int_0^{\infty} e^{-(s-a)t} dt = \left[\frac{e^{-(s-a)t}}{-(s-a)} \right]_0^{\infty} = \frac{1}{s-a}$$

2. $f(t) = 1$

Solution:

$$L(1) = \int_0^{\infty} e^{-st} dt = \left[\frac{e^{-st}}{-s} \right]_0^{\infty} = \frac{1}{s}$$

3. $f(t) = \cos(at)$

Solution:

$$L(\cos(at)) = \int_0^{\infty} e^{-st} \cos(at) dt = \left[\frac{e^{-st}}{a^2 + s^2} (-s \cos(at) + a \sin(at)) \right]_0^{\infty} = \frac{s}{a^2 + s^2}$$

4. $f(t) = \sin(at)$

Solution:

$$L(\sin(at)) = \int_0^{\infty} e^{-st} \sin(at) dt = \left[\frac{e^{-st}}{a^2 + s^2} (s \cos(at) - a \sin(at)) \right]_0^{\infty} = \frac{a}{a^2 + s^2}$$

5. $f(t) = \cosh(at)$

Solution:

$$L(\cosh(at)) = \int_0^{\infty} e^{-st} \cosh(at) dt = \int_0^{\infty} e^{-st} \left(\frac{e^{at} + e^{-at}}{2} \right) dt = \frac{s}{s^2 - a^2}$$

6. $f(t) = \sinh(at)$

Solution:

$$L(\sinh(at)) = \int_0^{\infty} e^{-st} \sinh(at) dt = \int_0^{\infty} e^{-st} \left(\frac{e^{at} - e^{-at}}{2} \right) dt = \frac{a}{s^2 - a^2}$$

7. $f(t) = t^n$

Solution:

$$L(t^n) = \int_0^{\infty} e^{-st} t^n dt = \left[t^n \frac{e^{-st}}{-s} \right]_0^{\infty} - \int_0^{\infty} \frac{e^{-st}}{-s} n t^{n-1} dt$$

$$\therefore L(t^n) = \frac{n}{s} L(t^{n-1}) = \frac{n}{s} \frac{n-1}{s} L(t^{n-2}) = \dots = \frac{n!}{s^{n+1}}$$

3.1.4 Linearity Property:

Laplace transform follows linearity property:

1. $L(f(t) \pm g(t)) = L(f(t)) \pm L(g(t))$
2. $L(cf(t)) = cL(f(t))$

Question: Find Laplace Transforms of the following functions:

1. $f(t) = \sin^2 t$

Solution:

$$L(\sin^2 t) = L\left(\frac{1 - \cos(2t)}{2}\right) = \frac{1}{2}L(1) - \frac{1}{2}L(\cos(2t)) = \frac{1}{2s} - \frac{1}{2}\left(\frac{s}{s^2 + 4}\right)$$

2. $f(t) = \cos^3 t$

Solution:

$$L(\cos^3 t) = L\left(\frac{3\cos(t) + \cos(3t)}{4}\right) = \frac{3}{4}L(\cos(t)) + \frac{1}{4}L(\cos(3t))$$

$$\therefore L(\cos^3 t) = \frac{3}{4}\left(\frac{s}{s^2 + 1}\right) + \frac{1}{4}\left(\frac{s}{s^2 + 9}\right)$$

3.1.5 Gamma Function:

1. $\Gamma(n+1) = \int_0^\infty e^{-t} t^n dt$
2. $\Gamma(n+1) = n\Gamma(n) = n!$
3. $\Gamma(1/2) = \sqrt{\pi}$
- 4.

$$L(t^n) = \frac{n!}{s^{n+1}} = \frac{\Gamma(n+1)}{s^{n+1}} \implies L(\sqrt{t}) = \frac{\Gamma(3/2)}{s^{3/2}} = \frac{\sqrt{\pi}}{2s^{3/2}}$$

3.2 First Shifting Theorem:

3.2.1 Theorem 1:

1. If $L(f(t)) = F(s)$, then $L(e^{at} f(t)) = F(s - a)$
2. **Proof:**

$$L(e^{at} f(t)) = \int_0^\infty e^{-st} e^{at} f(t) dt = \int_0^\infty e^{-(s-a)t} f(t) dt = F(s - a)$$

3. **Question:** Find $L(e^{-t} \sin(2t))$

Solution:

$$L(\sin(2t)) = \frac{2}{s^2 + 4} \implies L(e^{-t} \sin(2t)) = \frac{2}{(s+1)^2 + 4}$$

3.2.2 Theorem 2:

1. If $L(f(t)) = F(s)$, then $L(tf(t)) = -\frac{d}{ds}F(s)$

2. **Proof:**

$$\frac{d}{ds}F(s) = \int_0^\infty \frac{\partial}{\partial s}(e^{-st})f(t)dt = \int_0^\infty -te^{-st}f(t)dt = L(tf(t))$$

3. Hence,

$$L(t^n f(t)) = (-1)^n \frac{d^n}{ds^n} F(s)$$

4. **Question:** Find $L(te^{-2t} \sin(3t))$:

Solution:

$$\begin{aligned} L(\sin(3t)) &= \frac{3}{s^2 + 9} \\ \implies L(t \sin(3t)) &= -\frac{d}{ds} \left(\frac{3}{s^2 + 9} \right) = -\frac{3(-2s)}{(s^2 + 9)^2} = \frac{6s}{(s^2 + 9)^2} \\ \implies L(te^{-2t} \sin(3t)) &= \frac{6(s + 2)}{((s + 2)^2 + 9)^2} \end{aligned}$$

3.2.3 Theorem 3:

1. If $L(f(t)) = F(s)$ and $\lim_{t \rightarrow 0} \left[\frac{f(t)}{t} \right]$ exist, then $L\left(\frac{f(t)}{t}\right) = \int_s^\infty F(s)ds$

2. **Proof:**

$$\int_s^\infty F(s)ds = \int_0^\infty \left(\int_s^\infty e^{-st}ds \right) f(t)dt = L\left(\frac{f(t)}{t}\right)$$

3. Hence,

$$L\left(\frac{f(t)}{t^n}\right) = \int_s^\infty \int_s^\infty \dots (n - \text{times}) \int_s^\infty F(s)dsds \dots (n - \text{times})ds$$

3.3 Unit Step Functions:

1.

$$U(t - a) = \begin{cases} 0, & t < a \\ 1, & t \geq a \end{cases}$$

2.

$$\begin{aligned} L(U(t - a)) &= \int_0^\infty e^{-st}f(t)dt = \int_0^a 0dt + \int_a^\infty e^{-st}(1)dt \\ &= \int_a^\infty e^{-st}dt = \left[-\frac{e^{-st}}{s} \right]_a^\infty = \frac{e^{-as}}{s} \end{aligned}$$

3.3.1 Dirac Delta Function:

1.

$$\delta(t-a) = \begin{cases} \lim_{h \rightarrow 0} \left[\frac{1}{h} \right], & a < t < a+h \\ 0, & \text{otherwise} \end{cases}$$

2.

$$\begin{aligned} L(\delta(t-a)) &= \int_0^\infty \delta(t-a) e^{-st} dt = \int_a^{a+h} e^{-st} \lim_{h \rightarrow 0} \left[\frac{1}{h} \right] dt \\ &= \lim_{h \rightarrow 0} \left[\frac{1}{h} \left(\frac{e^{-st}}{-s} \right) \right]_a^{a+h} = \lim_{h \rightarrow 0} \left[\frac{e^{-(a+h)s} - e^{-as}}{-sh} \right] \\ &= \frac{e^{-as}}{s} \lim_{h \rightarrow 0} \left[\frac{1 - e^{-sh}}{h} \right] = \frac{e^{-as}}{s} \cdot 1 = e^{-as} \end{aligned}$$

Hence, $L(\delta(t-a)) = e^{-as}$

3.4 Second Shifting Theorem:

If $L(f(t)) = F(s)$, and $G(t) = \begin{cases} f(t-a), & t \geq a \\ 0, & t < a \end{cases}$ then, $L(G(t)) = e^{-as} F(s)$

Question: Find $L(G(t))$ for $G(t) = \begin{cases} \cos(t - 2\pi/3), & t \geq 2\pi/3 \\ 0, & t < 2\pi/3 \end{cases}$

Solution:

$$L(\cos(2t)) = \frac{s}{s^2 + 1} \implies L(G(t)) = e^{-2\pi s/3} \cdot \frac{s}{s^2 + 1}$$

3.5 Change of Scale Theorem:

If $L(f(t)) = F(s)$, then $L(f(at)) = \frac{1}{a} F\left(\frac{s}{a}\right)$

Proof:

$$L(f(at)) = \int_0^\infty e^{-st} f(at) dt = \int_0^\infty e^{-\frac{s}{a}(at)} f(at) \frac{d(at)}{a} = \frac{1}{a} F\left(\frac{s}{a}\right)$$

3.6 Laplace Transform of Derivatives:

$$L(f'(t)) = sF(s) - F(0)$$

$$L(t^n f(t)) = s^n F(s) - s^{n-1} F'(0) - \dots - F^{(n-1)}(0)$$

3.7 Laplace Transform of Integrals:

$$L\left(\int_0^t f(t)dt\right) = \frac{1}{s}F(s)$$

$$L\left(\int_0^t \int_0^t \dots (n - \text{times}) \int_0^t f(t)dt dt \dots (n - \text{times}) dt\right) = \frac{1}{s^n}F(s)$$

Question: Find $L\left(e^{-t} \int_0^{-t} \frac{\sin t}{t} dt\right)$:

Solution:

$$L(\sin t) = \frac{1}{s^2 + 1}$$

$$\Rightarrow L\left(\frac{\sin t}{t}\right) = \int_s^\infty \frac{1}{s^2 + 1} ds = \tan^{-1}s \Big|_s^\infty = \frac{\pi}{2} - \tan^{-1}s = \cot^{-1}s$$

$$\Rightarrow L\left(\int_0^t \frac{\sin t}{t} dt\right) = \frac{1}{s} \cot^{-1}s$$

$$\Rightarrow L\left(e^{-t} \int_0^t \frac{\sin t}{t} dt\right) = \frac{1}{s+1} \cot^{-1}(s+1)$$

3.8 Periodic Functions:

$$\int_0^\infty e^{-st} f(t) dt = \int_0^a e^{-st} f(t) dt + \int_a^{2a} e^{-st} f(t) dt + \int_{2a}^{3a} e^{-st} f(t) dt + \dots$$

In the second integral, let $t = T+a \Rightarrow t : [a, 2a] \rightarrow T : [0, a]$

Similarly, in the third integral, let $t = T+2a \Rightarrow t : [2a, 3a] \rightarrow T : [0, a]$

$$\int_0^\infty e^{-st} f(t) dt = \int_0^a e^{-st} f(t) dt + \int_0^a e^{-sT} f(T) dT + \int_0^a e^{-sT} f(T) dT + \dots$$

$$\therefore \int_0^\infty e^{-st} f(t) dt = (1 + e^{-as} + (e^{-as})^2 + \dots) \int_0^a e^{-st} f(t) dt$$

$$\therefore \int_0^\infty e^{-st} f(t) dt = (1 - e^{-as})^{-1} \int_0^a e^{-st} f(t) dt$$

3.9 Initial Value Theorem:

If $L(f(t)) = F(s)$, then

$$\lim_{t \rightarrow 0} f(t) = \lim_{s \rightarrow \infty} [sF(s)]$$

3.10 Final Value Theorem:

If $L(f(t)) = F(s)$, then

$$\lim_{t \rightarrow \infty} f(t) = \lim_{s \rightarrow 0} [sF(s)]$$

4 Inverse Laplace Transforms

4.1 Introduction:

$$L(f(t)) = F(s) \implies L^{-1}(F(s)) = f(t)$$

$$L^{-1}\left(\frac{1}{s}\right) = 1 \quad L^{-1}\left(\frac{1}{s-a}\right) = e^{at}$$

$$L^{-1}\left(\frac{a}{s^2 + a^2}\right) = \sin(at) \quad L^{-1}\left(\frac{s}{s^2 + a^2}\right) = \cos(at)$$

$$L^{-1}\left(\frac{a}{s^2 - a^2}\right) = \sinh(at) \quad L^{-1}\left(\frac{s}{s^2 - a^2}\right) = \cosh(at)$$

4.2 Properties:

4.2.1 First Shifting Property:

$$L^{-1}(F(s)) = f(t) \implies L^{-1}(F(s-a)) = e^{at}f(t)$$

Question: Find the Laplace Inverse for:

1.

$$L^{-1}\left(\frac{1}{(s+2)^2}\right) = e^{-2t}L^{-1}\left(\frac{1}{s^2}\right) = te^{-2t}$$

2.

$$L^{-1}\left(\frac{s}{(s+2)^2}\right) = L^{-1}\left(\frac{s+2}{(s+2)^2} - \frac{2}{(s+2)^2}\right) = e^{-2t}(1-2t)$$

4.2.2 Change of Scale Property:

$$L^{-1}(F(s)) = f(t) \implies L^{-1}(F(as)) = \frac{1}{a}f\left(\frac{t}{a}\right)$$

4.2.3 Differentiation Property:

$$L^{-1}(F(s)) = f(t) \implies L^{-1}(F'(s)) = -tf(t)$$

Question: Find the Laplace Inverse for:

$$L^{-1}\left(\frac{s}{(s^2 + a^2)^2}\right) = -\frac{1}{2}L^{-1}\left(\left(\frac{1}{s^2 + a^2}\right)'\right) = \frac{t}{2a}\sin(at)$$

4.2.4 Multiplication with s Property:

$$L^{-1}(F(s)) = f(t) \implies L^{-1}(sF(s)) = -\frac{d}{dt}f(t)$$

Question: Find the Laplace Inverse for:

$$\begin{aligned} L^{-1}\left(\frac{s}{(s^2+2)^2+4}\right) &= \frac{d}{dt}L^{-1}\left(\frac{1}{(s^2+2)^2+4}\right) \\ &= \frac{d}{dt}\left(e^{-2t}\frac{\sin(2t)}{2}\right) = e^{-2t}[\cos(2t) - \sin(2t)] \end{aligned}$$

4.2.5 Division with s Property:

$$L^{-1}(F(s)) = f(t) \implies L^{-1}\left(\frac{F(s)}{s}\right) = \int_0^t f(t)dt$$

Question: Find the Laplace Inverse for:

$$\begin{aligned} L^{-1}\left(\frac{1}{s(s^2-2s+5)}\right) &= \int_0^t L^{-1}\left(\frac{1}{(s-1)^2+4}\right)dt = \int_0^1 e^t \frac{\sin(2t)}{2}dt \\ &= \frac{1}{2} \left[\frac{e^t}{1+4}(\sin(2t) - 2\cos(2t)) \right] \Big|_0^t = \frac{e^t}{10}(\sin(2t) - 2\cos(2t)) + \frac{1}{5} \end{aligned}$$

4.2.6 Second Shifting Theorem:

$$L^{-1}(F(s)) = f(t) \implies L^{-1}(e^{as}F(s)) = f(t) \Big|_{t \rightarrow t-a} \cdot U(t-a)$$

Question: Find the Laplace Inverse for:

$$L^{-1}\left(\frac{e^{-s}}{s+2}\right) = U(t-1)L^{-1}\left(\frac{1}{s+2}\right) \Big|_{t \rightarrow t-1} = U(t-1)e^{-2(t-1)}$$

4.3 Convolution:

1. **Definition:** $f(t)*g(t) = \int_0^t f(u)g(t-u)du$
2. **Theorem:** $L(f(t)*g(t)) = L(f(t)) \cdot L(g(t)) = F(s) \cdot G(s)$
3. **Proof:**

$$\begin{aligned} L(f(t)*g(t)) &= \int_0^\infty e^{-st}(f(t)*g(t))dt = \int_0^\infty \int_0^t e^{-st}f(u)g(t-u)du \\ &= \int_0^\infty \int_u^0 e^{-st}f(u)g(t-u)dudt = \int_0^\infty \int_u^0 e^{-s(t-u)}e^{-su}f(u)g(t-u)dtdu \end{aligned}$$

Let $t - u = v \implies dt = dv$

$$= \int_0^\infty \int_0^\infty e^{-sv} e^{-su} f(u) g(v) dv du = \int_0^\infty e^{-su} f(u) du \cdot \int_0^\infty e^{-sv} g(v) dv$$

Hence,

$$L(f(t) * g(t)) = F(s) \cdot G(s)$$

4.

$$L^{-1}(F(s) \cdot G(s)) = f(t) * g(t) = \int_0^t f(u) g(t-u) du$$

5. **Question:** Find Laplace Inverse of:

$$\begin{aligned} L^{-1}\left(\frac{1}{(s-1)(s-2)}\right) &= L^{-1}\left(\frac{1}{(s-1)} \cdot \frac{1}{(s-2)}\right) = e^t * e^{2t} \\ &= \int_0^t e^u e^{2(t-u)} du = e^{2t} \int_0^t e^{-u} du = e^{2t} - e^t \end{aligned}$$

4.4 Applications of Laplace Transforms:

4.4.1 Solving an Ordinary Differential Equation:

1. $L(f'(t)) = sF(s) - f(0)$
2. $L(f''(t)) = s^2F(s) - sf(0) - f'(0)$

3. **Question:** Find $y(t)$: $y' - y = t$, $y(0) = 0$

Solution: Taking Laplace Transform on both sides,

$$\begin{aligned} L(y') - L(y) &= L(t) \implies sY(s) - y(0) - Y(s) = \frac{1}{s^2} \\ \implies (s-1)Y(s) &= \frac{1}{s^2} \implies Y(s) = \frac{1}{s^2(s-1)} \\ \implies y(t) &= L^{-1}\left(\frac{1}{s^2(s-1)}\right) = \int_0^t \int_0^t e^t dt dt = e^t - t - 1 \end{aligned}$$

4. **Question:** Find $y(t)$: $y'' - 4y' + 8y = e^{2t}$, $y(0) = 2$, $y'(0) = -2$

Solution: Taking Laplace Transform on both sides,

$$\begin{aligned} L(y'') - 4L(y') + 8L(y) &= L(e^{2t}) \\ \implies s^2Y(s) - sy(0) - y'(0) - 4Y(s) - 4y(0) + 8Y(s) &= \frac{1}{s-2} \\ \implies Y(s)(s^2 - 4s + 8) - 2s + 10 &= \frac{1}{s-2} \implies Y(s) = \frac{2s^2 - 14s + 21}{(s-2)(s^2 - 4s + 8)} \\ \implies y(t) &= L^{-1}\left(\frac{2s^2 - 14s + 21}{(s-2)(s^2 - 4s + 8)}\right) = \frac{e^{2t}}{4}(7\cos(2t) - 12\sin(2t) + 1) \end{aligned}$$

4.4.2 Solving an Integral Equation:

1. $L(\int_0^t y dt) = \frac{1}{s}F(s)$

2. **Question:** Find $y(t)$: $y + \int_0^t y dt = t^2 + 2t$

Solution: Taking Laplace Transform on both sides,

$$L(y) + L(\int_0^t y dt) = L(t^2 + 2t)$$

$$Y(s) + \frac{1}{s}Y(s) = \frac{2}{s^3} + \frac{2}{s^2} \implies Y(s) = \frac{2}{s^2}$$

$$\therefore y(t) = L^{-1}\left(\frac{2}{s^2}\right) = 2t$$

3. **Question:** Find $y(t)$: $y' + 2y + \int_0^t y dt = 2\cos t$, $y(0) = 1$

Solution: Taking Laplace Transform on both sides,

$$L(y') + 2L(y) + L(\int_0^t y dt) = L(2\cos t)$$

$$sY(s) - y(0) + 2Y(s) + \frac{1}{s}Y(s) = \frac{2s}{s^2 + 1}$$

$$\implies (s + 2 + \frac{1}{s})Y(s) = \frac{2s}{s^2 + 1} + 1 \implies Y(s) = \frac{s}{s^2 + 1}$$

$$\implies y(t) = L^{-1}\left(\frac{s}{s^2 + 1}\right) = \cos t$$

4. **Question:** Find $y(t)$: $y = 1 + 2 \int_0^t e^{-2u} y(t-u) du$

Solution: Taking Laplace Transform on both sides,

$$L(y) = L(1) + 2L\left(\int_0^t e^{-2u} y(t-u) du\right) = L(1) + 2L(e^{-2t} * y(t))$$

$$Y(s) = \frac{1}{s} + 2L\left(\frac{1}{s+2} \cdot Y(s)\right) \implies Y(s) = \frac{s+2}{s^2}$$

$$\implies y(t) = L^{-1}\left(\frac{1}{s} + \frac{2}{s^2}\right) = 1 + 2t$$

4.4.3 Solving a System of Linear Differential Equations:

1. **Question:** Solve the following system of linear differential equations:

$$\frac{dx}{dt} + y = \sin t, \quad \frac{dy}{dt} + x = \cos t, \quad x(0) = 2, y(0) = 0$$

Solution: Taking Laplace Transform on both sides,

$$sX(s) - x(0) + Y(s) = \frac{1}{s^2 + 1}, \quad sY(s) - y(0) + X(s) = \frac{s}{s^2 + 1}$$

$$sX(s) + Y(s) = \frac{1}{s^2 + 1} + 2, \quad X(s) + sY(s) = \frac{s}{s^2 + 1}$$

Multiplying equation 2 with s,

$$sX(s) + s^2Y(s) = \frac{s^2}{s^2 + 1}$$

Subtracting with equation 1,

$$Y(s)[s^2 - 1] = -\frac{s^2 + 3}{s^2 + 1} \implies Y(s) = -\frac{s^2 + 3}{(s^2 - 1)(s^2 + 1)}$$

$$Y(s) = -\frac{s^2 + 1 + 2}{(s^2 - 1)(s^2 + 1)} = -\frac{2}{s^2 - 1} + \frac{1}{s^2 + 1}$$

$$\therefore y(t) = \sin(t) - 2\sinh(t)$$

$$\text{Similarly, } x(t) = 2\cosh(t)$$

4.5 Solving a System of ODEs using Linear Algebra:

4.5.1 For Distinct Eigenvalues:

Let $y' = Ay$ where $y = [y_1, y_2, y_3, \dots, y_n]^T$ where $A = [a_{ij}]_{m \times n}$

If eigenvalues of A are $\lambda_1, \lambda_2, \lambda_3, \dots, \lambda_n$ and corresponding eigenvectors are $v_1, v_2, v_3, \dots, v_n$, then the general solution of the system of ODEs is:

$$y = c_1 v_1 e^{\lambda_1 t} + c_2 v_2 e^{\lambda_2 t} + c_3 v_3 e^{\lambda_3 t} + \dots + c_n v_n e^{\lambda_n t}$$

4.5.2 For 2x2 Matrix with Same Eigenvalues:

Let $y' = Ay$ where $A = [a_{ij}]_{2 \times 2}$

If eigenvalues are λ_1, λ_1 i.e., the geometric multiplicity can be 1 or 2.

1. If GM is 2:

then there exist 2 linearly independent eigenvectors (v_1, v_2)

$$y = c_1 v_1 e^{\lambda_1 t} + c_2 v_2 e^{\lambda_1 t}$$

2. If GM is 1:

then there exist only one linearly independent eigenvector v_1

$$(A - \lambda_1 I)v_1 = 0 \implies y_1 = c_1 v_1 e^{\lambda_1 t}$$

and the other eigenvector is v_2

$$(A - \lambda_1 I)v_2 = v_1 \implies y_2 = c_2(v_2 + v_1 t)e^{\lambda_1 t}$$

$$\therefore y = c_1 v_1 e^{\lambda_1 t} + c_2(v_2 + v_1 t)e^{\lambda_1 t}$$

4.5.3 For 3x3 Matrix with Same Eigenvalues:

Let $y' = Ay$ where $A = [a_{ij}]_{3 \times 3}$

If eigenvalues are $\lambda_1, \lambda_1, \lambda_1$ i.e., the geometric multiplicity can be 1 or 2 or 3.

1. If GM is 3:

then there exist 3 linearly independent eigenvectors (v_1, v_2, v_3)

$$y = c_1 v_1 e^{\lambda_1 t} + c_2 v_2 e^{\lambda_1 t} + c_3 v_3 e^{\lambda_1 t}$$

2. If GM is 2:

then there exist 2 linearly independent eigenvectors (v_1, v_2)

$$(A - \lambda_1 I)v_1 = 0 \implies y_1 = c_1 v_1 e^{\lambda_1 t}$$

$$(A - \lambda_1 I)v_2 = 0 \implies y_2 = c_2 v_2 e^{\lambda_1 t}$$

and then third eigenvector is v_3

$$(A - \lambda_1 I)v_3 = k_1 v_1 + k_2 v_2 \text{ where } k_1 \text{ and } k_2 \text{ are constants}$$

$$y_3 = c_3(v_3 + (k_1 v_1 + k_2 v_2)t)e^{\lambda_1 t}$$

$$\therefore y = c_1 v_1 e^{\lambda_1 t} + c_2 v_2 e^{\lambda_1 t} + c_3(v_3 + (k_1 v_1 + k_2 v_2)t)e^{\lambda_1 t}$$

3. If GM is 1:

then there exist only one linearly independent eigenvector v_1

$$(A - \lambda_1 I)v_1 = 0 \implies y_1 = c_1 v_1 e^{\lambda_1 t}$$

and the second eigenvector is v_2

$$(A - \lambda_1 I)v_2 = v_1 \implies y_2 = c_2(v_2 + v_1 t)e^{\lambda_1 t}$$

and the third eigenvector is v_3

$$(A - \lambda_1 I)v_3 = v_2 \implies y_3 = c_3(v_3 + v_2 t + v_1 t^2)e^{\lambda_1 t}$$

$$\therefore y = c_1 v_1 e^{\lambda_1 t} + c_2(v_2 + v_1 t)e^{\lambda_1 t} + c_3(v_3 + v_2 t + v_1 t^2)e^{\lambda_1 t}$$

5 Partial Differential Equations:

5.1 Separation of Variables:

1. **Question:** Solve: $u_{xx} - u_y = 0$

Solution: Let $u = X(x)Y(y)$

$$\implies u_{xx} = X''Y, \quad u_y = XY'$$

$$\implies X''Y - XY' = 0 \implies \frac{X''}{X} = \frac{Y'}{Y} = k$$

$$X'' = kX \implies m^2 = k \implies m = \pm\sqrt{k}$$

$$\implies X(x) = c_1 e^{\sqrt{k}x} + c_2 e^{-\sqrt{k}x}$$

$$Y' = kY \implies m = k \implies Y(y) = c_3 e^{ky}$$

$$\text{Hence, } u = (c_1 e^{\sqrt{k}x} + c_2 e^{-\sqrt{k}x}) c_3 e^{ky}$$

2. **Question:** Solve: $3u_x + 2u_y = 0$, $u(x, 0) = 4e^{-x}$

Solution: Let $u = X(x)Y(y)$

$$u_x = X'Y, \quad u_y = XY'$$

$$\implies 3X'Y + 2XY' = 0 \implies \frac{3X'}{X} = -\frac{2Y'}{Y} = k$$

$$3X' = kX \implies 3m = k \implies X(x) = c_1 e^{kx/3}$$

$$2Y' = -kY \implies 2m = -k \implies Y(y) = c_2 e^{-ky/2}$$

$$\text{Hence, } u = c_1 c_2 e^{kx/3} e^{-ky/2}$$

$$\text{But given, } u(x, 0) = 4e^{-x} \implies c_1 c_2 = 4, k = -3$$

$$\text{Hence, } u(x, y) = 4e^{-x} e^{3y/2}$$